



Battle of Britain Developers Group

BDG Patch 0.99 Manual

FOREWORD:	5
PART I: INSTALLATION GUIDE	6
1.1 BDG Installation	7
1.2 Record of Improvements from Original Rowan Battle of Britain	8
PART II: IN-GAME GUIDE	15
2.1 Configuring the game through the configuration screen	16
2.1.1 BDG Field of View Configuration Screen.....	16
2.1.1.1 Small Field of View.....	16
2.1.1.2 Medium Field of View	17
2.1.1.3 Large Field of View	17
2.1.1.4 Small-Toggle and Large-Toggle Values	17
2.1.1.5 Maximum and Minimum Field of Views	17
2.1.1.6 Initial Field of View	17
2.1.1.7 Padlock the target at the centre of the screen	17
2.1.2 BDG Game Configuration Screen	19
2.1.2.1 No Friendly Collisions	19
2.1.2.2 No Engine Cutout.....	19
2.1.2.3 No Stall Horn.....	19
2.1.2.4 Stragglers	20
2.1.2.5 Fighter Damage	20
2.1.2.6 Draw Padlock Centre Box	20
2.1.2.7 Enable Enemy Position Indicator.....	20
2.1.2.8 Play Intro Video.....	20
2.1.2.9 Skip Quick Videos	20
2.1.2.10 Skip Videos	20
2.1.2.11 Cont/Quit Quickmission.....	20
2.1.2.12 Cont/Quit Campaign.....	21
2.1.2.13 Head Bobbing	21
2.1.2.14 No padlock Head Bob	21
2.1.2.15 Preload Standard	21
2.1.2.16 Auto Preload	21
2.1.2.17 Fix Keyboard Rudder	21
2.1.2.18 High-Res Landscape.....	21
2.1.2.19 Enable AutoGen.....	22
2.1.2.20 TrackIR	22
2.1.2.21 Padlock overrides TrackIR	23
2.1.2.22 Near Clip Cockpit	23
2.1.2.23 Sky Poly Size Factor	23
2.1.2.24 EPI Radius/EPI Y Radius	24
2.1.2.25 Pan Speed Factor	24

2.1.2.26 Artificial Horizon Size	24
2.1.2.27 View Distance Factor	24
2.1.2.28 WK_Landscape.....	24
2.1.2.29 Debug Output.....	24
2.1.2.30 Gun Convergence	24
2.2 New Features configurable outside of the BDG UI	24
2.2.1 WinAmp music support	24
2.2.2 Leaning left and right out of the cockpit.....	25
2.2.3 Luftwaffe bomber formations and changes to their behaviour	25
2.2.4 Time deceleration	26
2.2.5 Trimming the aircraft	26
2.2.6 New Padlock Mode	26
2.2.7 Autopilot.....	26
2.2.8 Other Aircraft View.....	27
2.2.9 Labels	27
2.2.10 Track View Range.....	28
2.2.11 Peripheral Vision Range	28
2.2.12 Smoother Joystick Deadzone.....	28
2.2.13 Starting from Aircraft Bunkers/Pens	28
2.2.14 Gunsight and the Me-109.....	28
2.2.15 Keyboard Mapping.....	29
2.2.16 New Instrument View	29
2.2.17 Aircraft Independent Pilot Positions.....	29
2.2.18 Infoline Configuration	29
2.2.19 Roving Cam	30
2.2.20 UI Refresh.....	30
2.2.21 Belly landing	30
2.2.22 Bullet dispersion.....	30
2.2.23 High Speed Buffeting	31
2.2.24 Aero Surface Deflection	31
2.2.25 Changed aircraft collision effects	31
2.2.26 Frame-rate counter	31
2.3 New Keys and their function	31
PART III: THE CAMPAIGN	33
3.1 Campaign Code Changes.....	34
3.1.1 Missing He111 Statistics	34
3.1.2 Negative Aircraft Count.....	34
3.1.3 Losing the Campaign due to Excessive Convoy Losses	34
3.1.4 Wake Up Problem when Flying for a Specific Squadron	34
3.1.5 Reworking of All Campaign Statistical Counters.....	35
3.2 RAF Fighter Tactics	35
3.2.1 Readiness.....	35
3.2.2 Deployment.....	36
3.3 Procedures for Campaign Tasks.....	38
3.3.1 RAF Patrols	39
3.3.1.1 Patrols with Directives	39
3.3.1.2 Patrols set manually	40

3.3.1.3 Patrol Modifications	42
3.3.2 RAF Route Manipulation	47
3.3.3 Single Squadron Campaign	52
PART IV: DEVELOPERS VIEW	57
4.1 Is that the sound of thunder or me shooting holes into that innocent cloud?.....	58
4.2 Understanding the cardbase.rc file.....	58
4.3 AutoGen	60
4.3.1 AutoGen history and technical specifications	60
4.3.2 AutoGen implementation.....	61
4.4 BDG Code Bug Fixes.....	63
4.4.1 BoB crash fixes	63
4.4.2 Cloud Randomness	63
4.4.3 Keyboard Rudder Fix	64
4.4.4 Joystick Fix	64
4.4.5 Write Array Repair	64
4.4.6 Error message output.....	64
4.4.7 Panning bug.....	64
4.4.8 Gun and Cannon sounds	65
4.4.9 Flap buffet fix	65
4.4.10 Winds aloft correction.....	65
4.4.11 Ground turn or Radio turn bug	65
4.4.12 Fixed WK_LANDSCAPE 1 / Stutter fix	65
4.4.13 Stuka-grunt-bug fix.....	65
4.4.14 Pampa bug	66
4.4.15 Videos in Windows XP	66
4.4.16 Wrong files in 0.90 meant WinAmp did not work	66
4.4.17 The "no more hat panning after mouse pan" problem	66
4.4.18 Many bug fixes/improvements in robustness in 0.95	66
4.4.19 Fuelcock Fix.....	66
PART V: APPENDICES	67
4.1 Battle of Britain Development Group History	68
4.2 Battle of Britain Development Group	70
4.3 Acknowledgements.....	73
4.4 3rd Party Modifications.....	73
4.5 References and Historical Sources	73

Foreword:

This is a modified version of Rowan's 'Battle of Britain' (In Germany titled 'Adlertag'). It was created by the **BDG**, the **B**attle of Britain / **M**iG Alley **D**evelopment **G**roup, building on the code that Rowan published.

This is a work-in-progress. We have tested this patch extensively for prevalent bugs and crash prone code and have not found any; otherwise we would not be producing this patch for public use. We encourage any feedback, both positive and negative to the public 'Battle of Britain' forum at SimHQ (Please see the 'Links' section 4.3). We have not been able to test this on every possible system configuration, so if you encounter any problems, please let us know.

Should you have any suggestions relating to the new features, for example the new variables or default values, please post to the thread(s) about the beta. If you have other wishes, you are of course free to post to other threads at SimHQ, but we make no promises and think we have heard most wishes.

This manual is laid out in five basic areas.

Part I is the Quick Start and Installation Guide. This section will show you to install the BDG files.

Part II is the In-game guide. This section will cover all the configuration changes that are possible. This section will also explain the game changes.

Part III is the Campaign Procedures and Tactics guide. This section will show procedures for typical campaign requirements, it will also cover some strategies to maximize your enjoyment of the dynamic campaign.

Part IV is the Developers guide. This section enters the esoteric world of modders, coders, and magicians. It explains several important BoB files as understood and used by the BDG coders.

Part V is our appendix. It covers the history of the BDG, the members of the BDG, some important BoB websites and reference locations and the index.

Part I: Installation Guide



1.1 BDG Installation

There are two official Rowan patches for Battle of Britain available; neither is required for the BDG patch. If you have already patched your copy of the game with either of these two patches, it does not matter; installing the BDG patch over a new installation of Battle of Britain is not required. Our update will overwrite the data supplied in the Rowan patch, as well as, any previous BDG patch.

If you consistently have problems after installation please uninstall Battle of Britain. Ensure that all folders are deleted after uninstall process. Reinstall in the following order.

Rowan's Battle of Britain basic game
BDG 0.99 patch

The BDG patch will provide updated German, Spanish, and French files. We were unable to update the Italian and Russian language files, as we had no translator. We suggest that you do not have any third party skin, sound, or interface modifications installed until after the BDG patch.

The self-installer will ask you to verify the actual file path to Battle of Britain; if you have used a non-standard path name then you will have to use the "Browse" button to direct the installer to the correct directory.

After installation you can still use the default Battle of Britain shortcut to start the game. You should see a line in the first page of the user interface, below the "Quit" that shows this is a BDG product.

This manual can be found under the \Docs folder from the root Battle of Britain directory.

1.2 Record of Improvements from Original Rowan Battle of Britain

Changes to 099 (November 2004)

- Increased ground activity (see the \objectadds folder if you are interested in these. You can delete the files if you do not want them)
- A new Historical Quick Mission Scenario- The Raid on the Bristol Aero Engine Factory at Filton, Bristol, featuring over 200 aircraft.
- Further improvements to terrain and ground object colouring.
- Improvements to the BDG Advisor, using illustrative pictures. Thanks to Stickman.
- UK Maps with airfield locations marked. See the \docs folder. Thanks to MarkShot.
- OPTIONAL engine sounds. See the \OPTIONAL-NEW-ENGINE-SOUNDS folder.
- Various Bugfixes.

Changes to 098 (June 2004)

- New bdg.txt variable ENGINE_ALWAYS_RUNNING. If it is "ON", the engine runs when you get into the aircraft, even if you choose manual engine management
- Tweaked cumulus cloud rendering
- New Historical missions by Bader representing four key events in the Battle
- Possible to select different squadrons in Quickstart missions
- Fixed crash after mission delete
- Made code Win95 compatible again
- Fixed broken labels in Replay
- Fixed the Map bug with TrackIR
- VehicleManager, a huge internal change to allow new and/or tweaked aircraft models
- Increased maximum roving cam speed
- Added a Crashlog/Mini dump feature. This should help find the cause of crashes on users' computers
- The bdg.txt lines "OBJECT/ADD" now support a heading value
- Object Placement Mode added for 3D object addition.

v098 Artwork Updates (June 2004) (over v097)

- 1) *4 New Historical Quick Mission Scenarios, with historically authentic squadron/Staffel selection.*

*August 12th - The raid on Portsmouth and Ventnor
August 15th - Short Brothers Factory, Rochester
August 18th - Airfield Attacks- Kenley
August 18th - Airfield Attacks- South Coast*

- 2) *Lighter cockpit glass reflections (all planes)*
- 3) *Updated wheel textures from Stickman.*
- 4) *Weathering and detailing updates on RAF Fighters.*
- 5) *Muzak's Luftwaffe Fighter Flyby sounds*
- 6) *Updated std res 109 tail.*
- 7) *Various "OPTIONAL" folders. SEE THE ENCLOSED READMES.*
 - i) *109 High Res skins from Muzak (for users with higher memory video cards). Std RLM 71/02/65 and late RLM 74/75/65 schemes*
 - ii) *Luftwaffe bomber emblems from Redband*
 - iii) *Me110 emblems from Redband*
 - iv) *Ju87 emblems from Redband*
 - v) *Spitfire roundel options to match different squadron variants*
 - vi) *Late Battle schemes for Bf109/BF110. There was a greater use of mottling in the later battle periods. Skins by HeinzBaby*
 - vii) *Late Battle schemes for Hurricane. The Night/White undersides gave way to "Sky Type S" undersurfaces only.*
- 8) *Updated Trees from Maususer/Stickman.*
- 9) *A Rather nice Luftwaffe deployment map from Stickman (see /docs)*
- 10) *Updated v098 keyguide from Muzak/Stickman (see /docs)*
- 11) *Higher resolution landscape textures for users with >32M video cards.*
- 12) *New Cows from Redband to match the breeds that were more common 60 years ago (not modern Fresians!)*
- 13) *Updated Emil cockpit from Stickman/Muzak.*

14) Updated Luftwaffe emblems and lettercodes from Redband.

15) Updated terrain matching and colouring.

16) New Defiant Skin from HeinzBaby.

17) New Blenheim Skin from HeinzBaby.

18) Location correction to several RDF structures to match landscape.

19) Corrections to saturation levels on many skins.

20) Corrections to model bugs on the Hurricane around the canopy

Changes from 0.97 to 0.973, mainly bug fixes

- The texture corruption and the main cause for the CTDs has been fixed (it was one cause for both).
- Clouds in replay is fixed.
- After the waypoint error in the campaign, you should now be able to go on, although possibly with some waypoint artefacts.
- Mouse pan is now unlimited in external view and hits a stop in internal view.
- Hires screenies (see special help text in the folder "docs").
- Fixed a wrong assignment for a certain joystick button.

Changes to 097 (September 15 2003)

- *New Weather, including, among other things rain, lightning, better sun, and fixed sky poly problems*
- *Skinner's features including DDS texture support*
- *"Udet buoy" model, and a related quick mission*
- *campaign fixes by ElFanto*
- *Fix: start-from-pen with easy FM*
- *Fix: Fuelcock problem in 109 and Hurricane*
- *Belly landings*
- *New bullet ballistics*
- *Rendering fixes for Radeon video cards*
- *Fix the error messages/warnings in the debug window in MSVC*
- *Mouse-controllable roving cam*
- *New keyboard-configuration file handling*
- *Reworked Track IR code (but this should be transparent to the user)*
- *Obscurable AC-labels*
- *More fixes to the infoline*

Changes to 0.96 (May 16 2003)

- *Fixed bug that, on some systems, disabled sound if the movie was enabled.*
- *Fixed error in the LABEL unit.*
- *Fixed error that placed TrackIR debug output to the info line.*
- *Ensured Padlock Centre Screen function works with all input devices.*

- *Implemented new Dead Zone algorithm*
- *Implemented starting from aircraft pens in take-off quick mission*
- *Changed the Me-109 gunsight off-centre to a user defined option*
- *New Instrument Quick View*
- *Fixed Auto Canopy bug that opened the canopy at slow speed when you were in a dogfight. Now only opens when your gear is down.*
- *Developed short labels that remove the aircraft callsign*
- *Added a variable for the user to define the peripheral range*
- *Allows user to implement no ammunitions counters in aircraft, historically accurate*
- *New aircraft independent pilot eye position*
- *Incorrect labelling of views in the infoline has been corrected*
- *Added more feedback on user-option-changes through the infoline*

Changes to 0.95 Beta (February 02 2003)

- *Check hires landscape in the BDG/game GUI to get high-res landscape texture tiles. Needs a good graphics card, preferably with 64MB memory or more.*
- *TrackIR: native interface, gunner uses TrackIR for head movement and joystick or mouse for the gun, "Padlock Overrides TrackIR" setting in BDG / game GUI.*
- *Flak improvements: Better effect, better tracer, more than one Flak shape, new bullet.*
- *Tweaked fighter damage behaviour.*
- *Bdg.txt variable "NO_HEAD_BOBBING_WHILE_PADLOCK".*
- *Bdg.txt variable "WINAMP_FADES_OUT".*
- *Bdg.txt variable "TRACKVIEWRANGE".*
- *Added bdg.txt variables "CONTINUE_QUIT_CAMPAIGN" and "CONTINUE_QUIT_QUICK".*
- *No more automatic quitting of 3D (after bailing out etc), you have to use ALT X to leave 3D.*
- *Better "Dither values" should mean that more area types are dithered, if you have texture filtering set to "all".*
- *109 has aileron trim now.*
- *The user is now able to change the way the labels look (they get more transparent in the distance).*
- *Autopilot mode, for example to have a stable plane while taking screen shots.*
- *'Track other Aircraft' view.*
- *Moved seldom used BDG-Switches out of the BDG-UI, added others*
- ***Many** bug fixes/stability enhancements*

Changes to 0.90 Beta (September 27, 2002)

- *Disable Stall Horn switch*
- *Disable 0-G engine cut-out switch*
- *Enabled different gun sounds*
- *Added WinAmp support*
- *Bomber-Damage-Behaviour*
- *Additional bug fixes for German version*
- *Red box around Padlock Centre "Area of Interest"*
- *Enabled enhanced sun-glare in-cockpit*

- *New "head-behind-gunsight" view in Me-109*
- *Flap-buffet on ground bug removed*
- *Replay Icons smaller*
- *Incorrect wind bullet drift error corrected*
- *Removed "GroundTurn bug"*
- *Enabled Gun Convergence Setting*

Changes to 0.73 Alpha:

- *Introduced Padlock Centre Screen function*
- *No Friendly Collisions switch*
- *Implemented Automatic Preloading*
- *There are now language packs for: French, Spanish*
- *Added x8 Preloading*
- *Output of average frame rate (actually this was added in an intermediary release) added*

Changes to 0.72:

- *Fixed a problem with debug output.*
- *Fixed minor bdgst.txt problem.*
- *Experimental change to the Me109 oleo legs. Feedback appreciated*

Changes to 0.71:

- *Added Auto tree lines; see the Autogen.txt file.*

Changes from first official BDG Beta to 0.70:

- *Made leaning left/right sticky*
- *Array write bugs.*
- *Trim with 'radio alarm' effect and faster. Maximum trim value increased*
- *Fixed error messages.*
- *Size of virtual artificial horizon configurable*
- *Size of EPI configurable*
- *Added Disable Continue/Quit switch*
- *Added Pan speed factor*
- *Fixed pan speed bug*
- *Removed date check*
- *Added Lib3d.dll version check*
- *Added timing diagnostic output ('stutter debug-output')*
- *Preload 'standard' files, so that they do not need to be loaded during flight*
- *Implemented a new, reduced stutter landscape mode*

First official BDG 0.60 Beta Patch (November 10th, 2001)

- *Implemented Zoom stuff*
- *Introduced disable head bobbing switch*
- *Disable Enemy Position Indicator (EPI)*
- *Added Time deceleration*
- *Keyboard rudder fix.*
- *Added "Joystick Reinit" switch*

3D Object Placement in BoB world (June 2004)

It is now possible for users to add their own 3D objects to the BoB landscape. "Object Placement mode" allows you to place almost any object anywhere on the map. The main limitation is that the objects are only graphic objects, they do not fight, can not be damaged or are not subject to collisions. The aim is to place objects like trees, houses, hay stacks, churches, telegraph poles etc. to liven up the terrain. The objects are automatically placed on top of the terrain and you can rotate them around the vertical axis, IOW you can choose their heading. The placed objects are written into the bdg.txt and will automatically be loaded in future sessions.

To start placing, in bdg.txt, set OBJECT_PLACEMENT_MODE to ON with an ascii editor like notepad.

You should set FORCE_WINDOWED_MODE to ON to be able to look into the bin catalogue while BoB runs. Note that some people report that placing objects in fullscreen mode leads to errors.

If you have problems, switch the throttle to keyboard in the controller screen in BoB.

When in BoB, go into roving cam mode (**ALT-V**). You should look at the landscape in such a way that the position you want to place the object on is in the middle of the screen. In particular, you need to look downwards at the ground or you will receive a warning and the process will fail.

Say you want to place a tree besides a road crossing. Rove near the crossing and look down on it.

It should be visible in the middle of your screen.

Now press the key **1** (on the normal keyboard, not on the numpad). It will ask you for the bin number. You can use the Bin Catalogue to find the number of the object you want to place.

As examples, object 46 is a 3.5" AA gun, 38 is a mobile kitchen, and 62 is a group of trees. So, enter 62 for trees.

Please note that there is only the Blenheim available for ground based aircraft at the moment (it is the only one with its undercarriage deployed). Also, be careful with people (object BLOKE) as they contain a waving arm animation that causes an error on exiting BoB (though the game runs ok and leaves no residual issues that we can see).

After leaving the dialog with "Ok", the trees should now appear in the spot you chose. You can now use the numbers 2,3,4,5 to move it this way or that way or this other way or that other way .

To reduce the step, press **6**. To increase step size, use **7**. If you have a particularly low frame rate, some key presses may be lost, simply press them again.

Using the keys **[8]** and **[9]** you can turn it, this way round or that way.
You can only rotate around the vertical axis. Again, the stepsize can be altered with the **[6]** and **[7]** keys.

When you are finished, leave BoB and look into the bdg.txt . The new objects are listed at the end, in a special section of objects that were set this session. Obviously, after the next session they may be listed together with all other bdg.txt objects.

Once an object has been set and written into the bdg.txt changes can only be made by manually editing the relevant entry in the bdg.txt file. So, if you want to move or delete an object retrospectively then you have find the object line in the bdg.txt.

So, for example, this places some trees (object 62) at Kenley airfield at a rotation angle of -110 degrees.

```
OBJECT_ADD 28615881,33120443,-110.000000,62
```

If you changed it to

```
OBJECT_ADD 28615881,33120443,20.000000,347
```

you would have an ambulance (object 347), facing in a different direction. Simple 😊

To find which entry is the one you want, rove to the object and with **[I]** switch on coordinate display or press space (see roving cam description) and compare the coordinates you get with those of the objects in the bdg.txt.

Note: If you look in your Program Files\Rowan Software\Battle of Britain folder you should see a couple of important text files.

bdg.txt contains settings information and

Readme-Additional_Ground_Activity.txt contains a list of objects that we have already developed.

It is not fully comprehensive, so we look forward to user-made updates **J**, but it is a start. You will need to copy the contents of this file into your bdg.txt file if you wish to upgrade the activity in these sites.

Locations included are:

RAF Airfields: Tangmere, Biggin Hill, Debden, Kenley, Hawkinge, Manston, Hornchurch, Warmwell, Gravesend, Croydon, Ford, Eastchurch, Lympne.

Ports, Cities: Portsmouth, Sheerness, Portland, London.

Luftwaffe Airfields: Marck (Calais), Coquelles, Calais.

Part II: In-Game Guide



2.1 Configuring the game through the configuration screen

The BDG added additional setup areas. At the main screen click on either PC Config or Sim Config, you will then see a bar across the top of the user interface with several options, click on the one that says “BDG”. That will take you to the BDG Field of View configuration screen.

2.1.1 BDG Field of View Configuration Screen

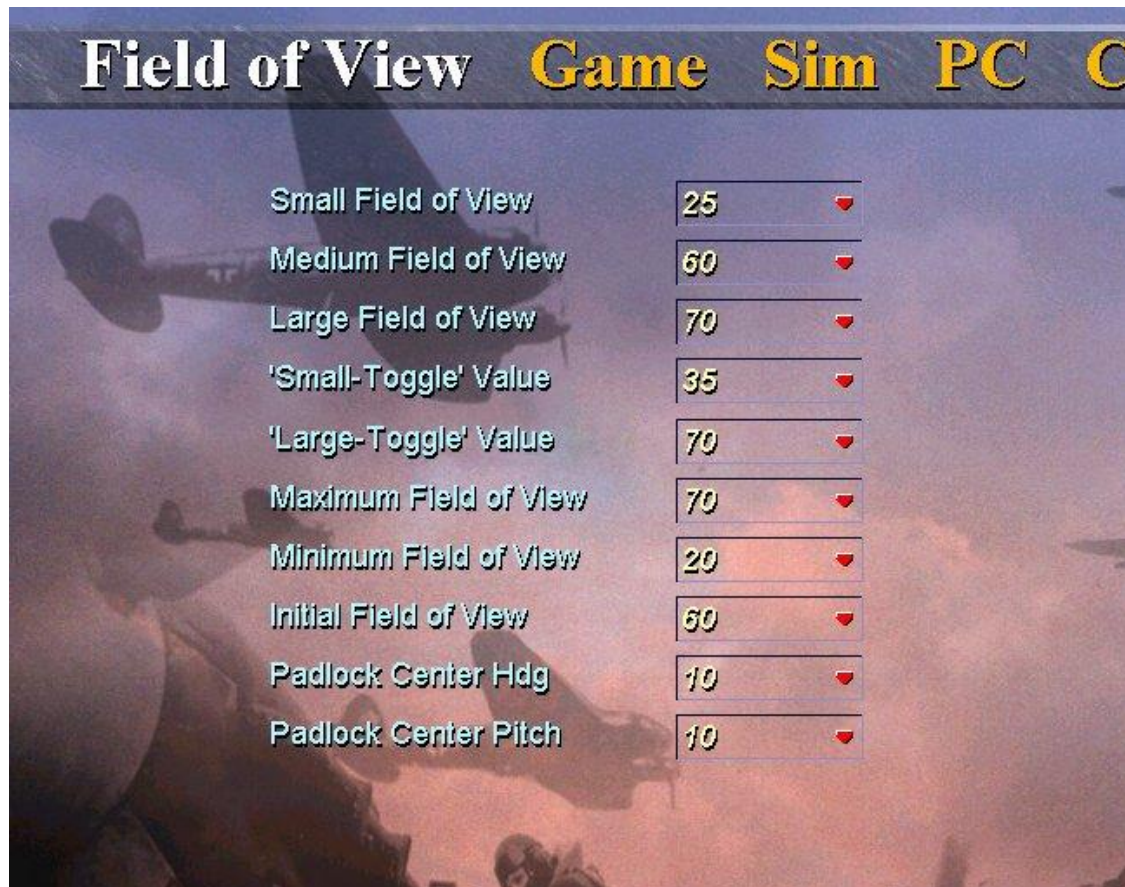


Figure 1: BDG User Interface for Field of View values

2.1.1.1 Small Field of View

Using the **SHIFT F9** key, you can focus your cockpit view to this preset value. This value is a slice of the normal cockpit view. Basically, the value you enter, (the default is 25 degrees) is the degrees around the centreline of your aircraft that your view is focused to. This view is extremely useful for those pilots who do not use aircraft text labels in order to “zoom” their eyesight and to make an aircraft identification.

2.1.1.2 Medium Field of View

Accessed using the **[SHIFT] [F10]** key, this is the same function as **Small Field of View**, only it accesses a middle cockpit view. Consider this the default cockpit view, especially if this value matches the **Initial Field of View**.

2.1.1.3 Large Field of View

Accessed using the **[SHIFT] [F11]** key, this has the same function as **Small Field of View** only with as much of the cockpit displayed as possible. This view can lead to problems if the number becomes too large. As the **Large Field of View** value gets larger (default is 70 degrees) the program has to force all the objects on to the screen. This may cause a “fisheye” view where objects in the middle of the screen are larger than normal, and objects at the edges of the screen are smaller. If this happens and you do not like the effect, simply lower this value.

2.1.1.4 Small-Toggle and Large-Toggle Values

Use the **[SHIFT] [F12]** key to toggle between these field of view values.

2.1.1.5 Maximum and Minimum Field of Views

Using the standard Rowan keys of **[CTRL](keypad)[-]** and **[CTRL](keypad)[+]**, these are the farthest zoom these keys can access. If you start at the most zoomed default value of 20 and hit the **[CTRL](keypad)[+]**, you can access every number from 20 to the default maximum of 70. In addition, the **[SHIFT] [F8]** key has been added, that will zoom into the smallest value and then zoom out to the largest and back again as long as you are holding the **[SHIFT] [F8]** key.

2.1.1.6 Initial Field of View

This is the cockpit field of view value you will see when you first enter the cockpit.

2.1.1.7 Padlock the target at the centre of the screen

You're flying on a banking left hand dive and you see a ME-109 that you can easily bounce. You quickly hit **[F1]** to padlock this aircraft and suddenly your view swings 120 degrees to the right to some Ju-88 that is two miles away and 5,000 feet above your altitude. Well, no more! Padlocking the target in the middle of your screen has been implemented. Its use is quite easy and activated with the single keystroke of **[S]**. You will see a red box outlining the padlock area; if the aircraft is in that box then it will be padlocked when the **[S]** key is released. It is not recommended using this function if there is more than one aircraft in the outlined area, if you pick up the wrong one, you will

have to hit the **[Esc]** key to return to the non-locked cockpit view then hit the **[S]** key a second time.

Keeping with the variability of all options, the area in the centre of the screen that is used to determine which target will be padlocked is user defined. If you enter the BDG setup screen under "Field of View" you will see two lines labelled, "**Padlock Centre Hdg**" and "**Padlock Centre Pitch**".

The **Padlock Centre Hdg** (Heading) is a wedge that extends horizontally from the centre of your current view. So if the value is '10', there will be a wedge of 20 degrees off your current heading in that the "Padlock Centre" will work

The **Padlock Centre Pitch** is a wedge that extends vertically from the centre of your current view. If that value is '10', there will be a wedge that extends 20 degrees from the plane your wings form, in the direction of your lift line.

Combining these two values will create a "padlock" pyramid that extends out from the centre of your current view. With the two variables you can manipulate the size of that cone. For example, if you make the "**Padlock Centre Pitch**" larger than the heading's value you create a rectangle that will pick up targets along the lift line as your pulling into them. To use this function in both the mouse and keyboard panning modes, you must hit the **[CTRL] [S]** button. This will realign your view to the forward view and you should no longer use the panning method you were using before. Example: You want to go from mouse panning to keyboard panning, hit **[CTRL] [S]**. You will go to the forward view and you can now use the keyboard for panning and the padlock centre will work. The "PC_USING_KEYBOARD" line in the *bdg.txt* file can be manually changed to set a default for this value. If you are using TrackIR, you need not do this as both mouse and keyboard input are neglected.



2.1.2 BDG Game Configuration Screen

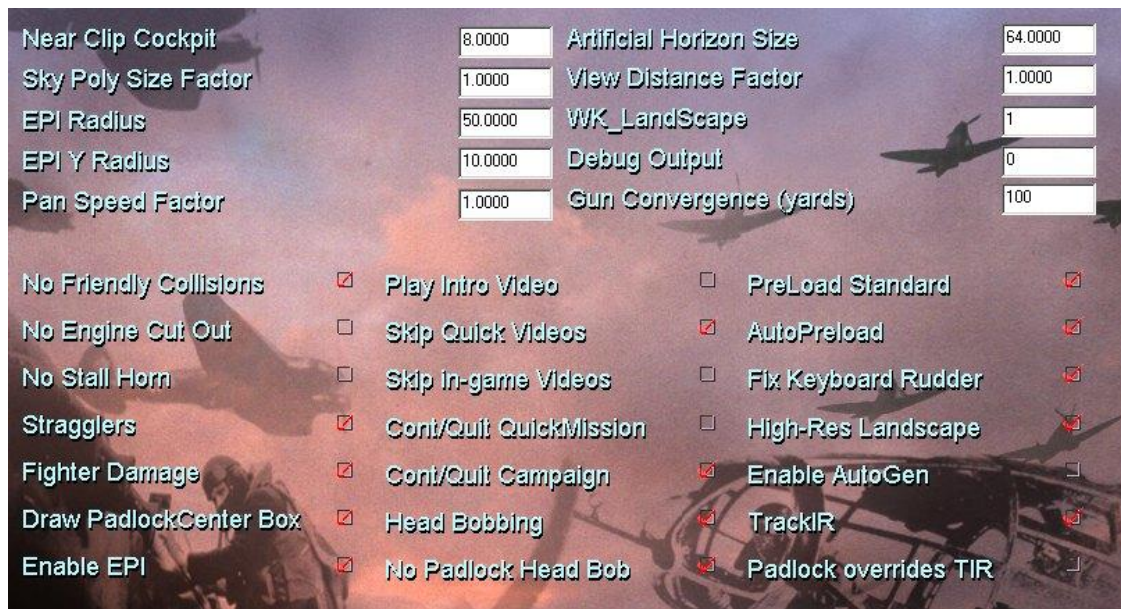


Figure 2: BDG User Interface for Game Toggles and Values (cropped)

There are 21 total toggles, or “ON/OFF” values in the BDG user interface. A red checkmark shows that the option is activated. To get help on what a certain option does, hover with the mouse cursor over one of the checkboxes and a short explanation should appear.

2.1.2.1 No Friendly Collisions

Ever been frustrated when you make that initial break into an enemy formation and you collide with one of your wingman? Well until now the only way to avoid this was to turn off mid-air collisions, resulting in less realism. Now you can activate the no friendly collision, while leaving on the Rowan switch for mid-air collisions. This will prevent you from taking damage from a friendly aircraft; yet still kill yourself when you impact that Ju-88 at a combined speed of 700 knots. Please note that you are still vulnerable to your wingmen's` bullets!

2.1.2.2 No Engine Cutout

Enable/Disables the engine cutout British fighters experienced when they entered a negative G dive. This feature only works if the complex engine management is enabled in the Rowan configuration screen.

2.1.2.3 No Stall Horn

Activating this checkbox disables the warning stall tone that is played in Spitfires when flying with less than 25% power. The BDG does not take any responsibility in case you forget to lower your gear before landing!

2.1.2.4 Stragglers

This option enables the new and varied Luftwaffe bomber damage. Please refer to section 2.2.3 for more details.

2.1.2.5 Fighter Damage

When activated, the fighter-AI will be likely to bail out if their engines are badly damaged. In case of the two-seaters, elimination of the rear-gunner will encourage the pilot to bail even sooner...

2.1.2.6 Draw Padlock Centre Box

This option disables the red rectangular outline that is visible when the **[S]** key is pressed to activate the **Padlock Centre Screen** function

2.1.2.7 Enable Enemy Position Indicator

This toggle shows enemy position indicator (EPI) on the upper left of the 3d world screen. This switch is independent of the in-game show HUD key. With this you can disable the EPI, but still have the attitude indicator showing in the upper right of the 3d screen. Please note you can not have the EPI while disabling the attitude indicator.

2.1.2.8 Play Intro Video

Enable/Disable the initial Battle of Britain video.

2.1.2.9 Skip Quick Videos

Enable/Disable Quick Mission Videos.

2.1.2.10 Skip Videos

Enable/Disable campaign videos.

2.1.2.11 Cont/Quit Quickmission

When you land your aircraft after a mission, there is a window that would pop-up asking you to continue or quit. A checkmark here means the window will appear.

"CONTINUE_QUIT_BOX_CAMPAIGN" and "CONTINUE_QUIT_BOX_QUICK" replace "DISABLE_CONTINUE_QUIT" (an old variable from 0.90), so your old setting from 0.90 will be ignored.

The two new ones let you disable them selectively, as the box proved to be quite useful in campaigns.

2.1.2.12 Cont/Quit Campaign

See previous (2.1.2.11). This checkbox governs the appearance of the box in campaign missions.

2.1.2.13 Head Bobbing

If this option is enabled, your “virtual” head will move in response to the G-forces that you’re pulling with your aircraft. Your screen will move with no panning inputs from you, this feature works independently of the Blackout/Whiteout/Redout effects your pilot may experience.

2.1.2.14 No padlock Head Bob

If this is selected, there will be no head bobbing while you are in padlock.

2.1.2.15 Preload Standard

Enables program to load standard *.bin* files during mission generation. This removes some of the stutters seen during action when these files are loaded.

2.1.2.16 Auto Preload

When you fly a quick mission for the first time, the computer remembers what files it was required to load during the game play and writes this information into a file. The next time you fly the same mission, these files are preloaded. This removes the “hiccup” or stutter that occurs when the file would be loaded during your flight in 3D. The whole process is totally automatic and currently only works with quick missions. We recommend setting this to “ON”.

2.1.2.17 Fix Keyboard Rudder

Fixes an error to allow the keyboard to fully deflect the rudder to the same degree a joystick would be able to. If you use rudder input from a joystick, rudder pedals, or twist grip, do not enable this feature.

2.1.2.18 High-Res Landscape

This option needs a good graphics card, preferably with 64MB memory or more. You will also have to switch “WK_LANDSCAPE” mode to ‘1’, so that you do not get excessive stutter or fps slowdown. The effect is most notably in cities and where you see jaggies, for example at the border of the runway. Since things are sharper, you will probably also see more “shimmering”. To reduce this, use anisotropic filtering, this option can be enabled in the settings of your video card.

2.1.2.19 Enable AutoGen

Enabling this option will start the Auto Generation of objects in the 3D world. It will replace the "Raised" terrain option in the Rowan setup page. Ensure that "Raised" terrain is disabled. For more information on AutoGen and how to setup the 3D objects please refer to section 3.3.

2.1.2.20 TrackIR

TrackIR is a hardware device that picks up your physical head movements and translates them into movements of your "virtual" head, so if you do not have one of these devices sit on your monitor, you can safely skip this (and the following) chapter.

BoB supports both normal "mouse emulation" and "native" mode of the TrackIR device, it is however strongly suggested to use the "native" mode.

There are a few things you **have** to do to make the native TrackIR -Interface work:

- Download and install version 1.4 (or later, you best use the most recent version) of the TrackIR interface software (which can be obtained from <http://www.naturalpoint.com>)
- Start the TrackIR interface each time before you start BoB.
- Enable the "native" mode in the TrackIR window
- Enable TrackIR in the PC Config -> BDG -> Game settings or in the *bdg.txt* directly.

At first you will not see much difference between mouse emulation mode and the native implementation using absolute coordinates. However, in native mode, if you "bang" your head against the 6 o'clock position and move back, or when you leaving your place (to get some beer etc.) and come back, you will not need to recenter the device.

If you play as pilot and use the native TrackIR mode, the mouse is unused (but can of course be mapped to the "view pan" – you just won't be able to adjust the view while TrackIR is active). However it is NOT recommended setting the interactive cockpit to the "unshifted" mouse! (As it will activate itself as soon as the mouse is moved even a tiny amount, and can not be deactivated.)

If you play as gunner, you have two separate inputs:

- TrackIR moves the view direction (the head).
- Either the joystick or the mouse moves the gun.

Even when playing with TrackIR, you can still "mark" things with the padlock functions like "padlock centre", "padlock enemy", "padlock friendly", "padlock ground object" etc.. If you have the target diamond switched on, the padlocked object will be marked with this symbol. So, for example, you could mark a plane that you think you damaged, but that does not even smoke. Even in the chaos of a big melee, the diamond will always show you the chosen aircraft.

You can either let padlock move your head - just like padlock does in all non-TrackIR simulations - or you can move the head with TrackIR and visually search for things. You

can switch between these two possibilities in the BDG -> game settings screen (please see the next chapter).

One note for users of NaturalPoint 2.0 software: The software defaults to “mouse look” mode and you must press **[F8]** once to set it to “enhanced mode” before starting BoB.

Please note that the NaturalPoint software by default “blocks” the **[F7]**, **[F8]**, **[F9]** and **[F12]** keys, so either you deactivate these keys in the NaturalPoint window or you change either the BoB or NaturalPoint key mappings.

2.1.2.21 Padlock overrides TrackIR

This option governs if the padlock will override your TrackIR view-input. If it is not checked, you can still use the padlock keys to mark objects with the targeting diamond, but you are free to pan around using the TrackIR. Please note that many wingman commands require the subject of the message to be marked by the targeting diamond (e.g. the “Help” command will make your wingmen attack your marked target only).

2.1.2.22 Near Clip Cockpit

This option may help with two possible problems in the cockpit:

When you have black bands in the instrument panel where the background of the instrument panel occludes the instruments, first you should enable `ALWAYS_USE_Z` in the *cardbase.rc* file as described in section 3.2. Only when that fails should you try to increase Near Clip Cockpit, maybe by 50% or more.

When you have no black bands but things in the cockpit (for example the headrest) get clipped off in large FoVs, then you should decrease Near Clip Cockpit. The value is, AFAIK, the near clip plane distance in decimetres. Rowan's value was 10.

For example, when you select Quickshot->basic->take off and use the mouse to look back while zoomed out (**[SHIFT]** **[F11]**) in the spitfire. With Near Clip Cockpit set to '10' (Rowan's value), you have definite problems (see the head rest), with '9' the problems have almost gone and with '8' they are gone at least as long as you don't have head movement due to heavy G's.

2.1.2.23 Sky Poly Size Factor

This variable determines the width of the sky box polygon. You can see the Sky Poly when you set **Sky Poly Size Factor** to, for example, '1' and choose a really large Field of View like '110' (degrees).

We are not really able to influence the height of the sky box yet. With the Sky Poly Size Factor you can influence the width. The default value is '1' in the Rowan code and this means that for large field of views (that Rowan's exe didn't allow), you will see the edge of the sky, '0' will extend the lighter lower sky. The beta testers have found that entering a value of '0.0000' into this field produces the best visual effects with the least amount of frame loss.

2.1.2.24 EPI Radius/EPI Y Radius

These two values change the size of the Enemy Position Indicator. The default ratio of Radius to Y Radius is 5 to 1. To keep the same default orientation we suggest you keep to this ratio. The numbers are in pixels.

2.1.2.25 Pan Speed Factor

Determines the panning-speed in the 3d world (this includes the 'photo' screen in the campaign!) Default value is '1', so a value of '2' would double the regular panning speed. It also affects the speed in death views.

2.1.2.26 Artificial Horizon Size

This setting determines the size of the Artificial Horizon. Default value is 50 (pixels).

2.1.2.27 View Distance Factor

As much as it pains us to admit it, this setting still has an unknown in-game effect. The default Rowan value is '1'. The beta testers have found that the best value is '0.50000'. This combined with the a Sky Poly Box value of '0.0000' and an Horizon Setting of "far" produces the best possible visual outcome.

2.1.2.28 WK_Landscape

This variable determines how landscape textures are drawn in the 3d world. While the default Rowan value is '0', values of '1' and '2' do normally reduce stutter. '2' activates experimental code and may lead to terrain graphic problems. For most people, mode '1' is the best mode, especially if the hi-res landscape is enabled.

2.1.2.29 Debug Output

This variable tells the programme to send output to a file used for debugging. Do not change it unless asked by a (*cue drum roll*) BDG coder. This option can slow your game down tremendously.

2.1.2.30 Gun Convergence

Permits you to set the distance where your bullet streams will converge. Default distance is 250 yards.

2.2 New Features configurable outside of the BDG UI

2.2.1 WinAmp music support

Battle of Britain originally was unable to play music, and to correct this, WinAmp support has been added. To use WinAmp support, start WinAmp **before** you start Battle of Britain and open a playlist. While in the 3D world you can then use the **CTRL** **F9** through **F12** keys to control the music.

CTRL	F9	<i>Previous Track</i>
CTRL	F10	<i>Start music</i>
CTRL	F11	<i>Stop music</i>
CTRL	F12	<i>Next Track</i>
ALT	+	<i>Increase music volume</i>
ALT	-	<i>Decrease music volume</i>

In previous BDG versions, WinAmp stopped playing when exiting the 3d. Setting the bdg.txt variable "WINAMP_FADES_OUT" to "OFF" means WinAmp will continue playing; you will however not be able to use the provided "WinAmp-control" key presses (**ALT** **Tab** to the WinAmp window instead).

Versions 0.95 and up also provide more feedback on WinAmp operations, i.e. if no WinAmp window is found in the background, and the user presses "PLAY" (**CTRL** **F10**), a "WinAmp window not found" message will appear in the infoline.

Another note on the WinAmp support... it will only work with versions lower than 3, as there seems to be no documentation on interaction with WinAmp 3.0 around (as it is being written right now). We hope to reinstate support for this version at a latter time. There is however a plug-in for WinAmp3 which enables BDG WinAmp functionality, which can be downloaded at Retro's website, see Section 4.3 for link.

2.2.2 Leaning left and right out of the cockpit

Although the original code allowed you to lean out of your cockpit, the BDG has made several changes to that function. The **Q** and **W** keys are now sticky, meaning that you press the key once, and you will remain in that position until you press the key a second time to return to the centre view. While using the **Q** and **W** views, you may now pan around the inside of the front half of that sphere. You can not look backward however, as the 3D model has been optimized and all you would see is air. If you have the "Offset Me-109 gun sight feature" enabled (by setting the line "ALWAYS_BEHIND_GUNSIGHT" in the *bdg.txt* to "ON"), you will hit **W** once to look behind the gun sight and **W** again to lean out of the right of the cockpit. If the Me-109 gun sight feature is not enabled, the **Q** and **W** move your view from behind the gunsight to the left and right sides, respectively, just like in the basic BoB game.

2.2.3 Luftwaffe bomber formations and changes to their behaviour

To facilitate large bomber formations, Rowan used a fairly simple bomber damage model. Bomber damage seemed to be either "fine and in formation", or "destroyed and falling to the ground". Now common sense tells us that a bomber with only one good engine and the second on fire or feathered will not be able to keep up with the rest of the bomber formation. This feature is now implemented in the BDG code, due to Retro's efforts.

Several outcomes are now possible when attacking bomber streams. It is now possible to kill and wound the gunners in the bombers, including the Ju-87 and the Me-110. If a bomber has an engine start to smoke due to damage, there is now a chance that it will pull out of formation and try to make it back home. Every bomber will not do this and I

have even seen the bombers stay in formation and put the engine fire out. All in all, however, bomber behaviour is more realistic.

2.2.4 Time deceleration

Just as you are able to accelerate time in the cockpit by using the **Tab** key, you are now able to slow down time in the 3D world by hitting the **SHIFT F7** key. There are some problems with this, however. When you look at the ground it will appear to move around in a random fashion. It will move normally for a second, stop, then try and catch up. Aircraft also exhibit this phenomenon as well. Still it will allow a quick look to gain that precious amount of Situational Awareness!

2.2.5 Trimming the aircraft

While the keys to trim the aircraft have not changed, their function has. To increase trim you no longer have to continuously hit the key, just press and hold down the key. When you press the key, the virtual "trim wheel" will start moving slowly, so that you are able to make small, accurate changes. The longer you hold the keyboard button down, the faster it will rotate (not unlike an alarm clock), this will allow you to make large inputs quickly. The maximum trim value has also been increased to allow you to trim the aircraft in any speed and configuration. This should make getting those damaged birds home easier.

2.2.6 New Padlock Mode

There is a new padlock mode that should remove the strange "upside-down" view you get when diving after an aircraft. As hardly any of Retro's changes come without a drawback, you will see a video stutter problem on rolling your aircraft with this feature activated. To enable this feature, open the *bdg.txt* file with any ASCII reader, like Notepad. Go to the line "BOB_PADLOCKFIX", and replace "OFF" with "ON". Save the file and when you start the game, this feature will be enabled.

2.2.7 Autopilot

Retro enabled an autopilot that however has a distinctive 'beta' feel to it.

CTRL A will toggle the autopilot, this hands the control over the player's AC over to the AI. It is able to follow waypoints and even fight, some unsolved problems made it however necessary to force the player into outside view while the autopilot is active. Changing the view back into the AC will deactivate the autopilot. Feedback on the autopilot's status is provided in the infoline.

Issues:

While the AP is also able to start engines and takeoff, it's somehow unable to raise or lower its gear.

Another unsolved problem is that if (with activated AP) the user goes to the map-screen and leaves, the engine-sound stops. Deactivating the AP does however get it back.

In previous versions, the keys **A**, **SHIFT A**, **CTRL A** and **ALT A** executed the 'any bandits' call. This has been moved to the **A** key only.

2.2.8 Other Aircraft View

CTRL I enters this view, which cycles the focus of your current view through all AI planes within a 10-kilometer radius around your plane. If you had an aircraft padlocked, pressing this keystroke will take you to this AC first, repeated presses cycle through the others. It might be a good idea activating the autopilot first.

2.2.9 Labels

We added quite a few features to the label functions: The user can now configure both the colour and the range where labels appear (sadly not through the BDG-GUI, but only in the *bdg.txt* ASCII file). The relevant commands are

```
LABELCOLOUR_FRIENDLY = 0000ff
LABELCOLOUR_ENEMY = 00ff00
```

Those govern the colour of the labels, their values are in RGB-hex format, meaning that the first 2 characters set the red-component of the colour (0 to 255), the next two characters govern the green part and the last two the blue part.

The values above are the standard ones, blue for allies, green for enemies. Leading zeroes may be omitted (0000ff is equal to ff). 000000 is black, ffffff is white.

Range is set through the next 3 lines:

```
LABEL_FULL_ALPHA = 2000
LABEL_HALF_ALPHA = 4000
LABEL_ZERO_ALPHA = 6000
```

'Alpha' means the 'opacity' of the colour, if alpha is zero the colour is effectively invisible. Full_Alpha is the maximum range till labels will be drawn 'fully', in ranges between FULL_ALPHA and HALF_ALPHA the alpha will decrease linear from 'full' to 'half' (i.e. 128), in ranges between HALF_ALPHA and ZERO_ALPHA it will decrease linear from 128 to 0; Aircraft that are farther away than the ZERO_ALPHA range do not get labels drawn at all. Units of those values are meters.

The changes to the label-system only apply to the single player game; Multiplayer stays the same as in "stock" BoB, save for a quick fix: some labels were drawn in cyan, which was actually a bug since Rowan intended them to be drawn in yellow.

The fading labels are activated by default, if you however want to go back to the "stock" Rowan labels, set the *bdg.txt* line "FADING_LABELS" to "OFF".

A new feature that was added with version 0.97 is that labels are now obscured by the cockpit. This also applies to the targeting diamond.

If you are bothered by the overly long labels, depicting both callsign and call number of all friendly aircraft, then you can reduce them to just the plane type by activating the

"SHORTENED_LABELS" variable in the *bdg.txt*. Aircraft in your squadron will however still display both callsign and call number.

2.2.10 Track View Range

The decimal value "TRACKVIEWRANGE" governs the initial range of the outside (track) view... this is also the initial value when the **[Esc]** key is pressed... sadly we are not sure about the units of this value, so a bit fiddling is in order.

2.2.11 Peripheral Vision Range

Another one of the situational awareness aids Rowan implemented are the dots that float on the edge of the screen, simulating the pilot's peripheral vision. In earlier versions your super God-like vision could track aircraft out of the corner of your eye for up to four kilometres... now this range is user controlled. The *bdg.txt* file has a variable called "PERIPHERAL_VISION_RANGE". In order to change it you must open the *bdg.txt* with Notepad or another text editing program and edit the value currently there. The default value is 4000, the distance is in meters. There is no corresponding field in the BDG GUI as of yet

2.2.12 Smoother Joystick Deadzone

A new coder, Budeye1, burst onto the scene and one of his fixes was to create a smoother transition from the centre point of the dead zone to the maximum deflection point on your joystick. If you have no dead zone on your BoB game configuration then activating this feature may not give you any benefit. For those that do, this feature should make your aircraft movements smoother and easier to control. It is defaulted to be "ON", in order to change it, open the *bdg.txt* and find the "BOB_SMOOTHER_DEADZONE" line; then change it from "ON" to "OFF".

2.2.13 Starting from Aircraft Bunkers/Pens

Realistically, when a pilot began a mission, he started from the pen the aircraft was moved to after flight. These aircraft pens were designed to support the aircraft and add to their survivability during an air raid. Now you can start from this position instead of the open field. This feature only works with the quick mission "*Basic/Take-off*", and if you choose an airfield with no aircraft pens you will start near hangers or trees. The default value for this feature is "ON", to change this variable, open the *bdg.txt*, find "START_FROM_PEN" and change it from "ON" to "OFF".

2.2.14 Gunsight and the Me-109

Starting with BDG patch 0.95, the Me-109 gunsight was historically modelled in its off-centre position. This required the pilot to shift slightly to the right in order to use the gunsight during a dogfight. The BDG modelled this and now requires the user to hit the "*Lean Right*" key (**[W]**) once to "get behind the gunsight". If you are bothered by this feature, you can deactivate it by opening the *bdg.txt*, and changing the line "ALWAYS_BEHIND_GUNSIGHT" from "OFF" to "ON".

2.2.15 Keyboard Mapping

Setting the "GET_KEYMAPS_FROM_FILE" variable to "ON" enables reading of command from the file "keys.xml" located in the /keyboard directory. You will be able to edit this file with the ordinary notepad application; it may be advisable though using the BDG keyboard mapper which can be found on the BDG website.

If you have enabled the above mentioned *bdg.txt* switch and do not have a "keys.xml" yet, it will get created automatically, so if you corrupt it somehow, just delete it. Additionally, the program will check for wrong syntax and false commands or keys, if any errors are detected, a small log will be created and placed into the "err.txt" file in the same directory.

2.2.16 New Instrument View

There is now a quick view button to "glance" at the instruments and then return to where you were looking; the default key is the  key on the keypad.

2.2.17 Aircraft Independent Pilot Positions

These variables will allow for differing pilots eye positions for each flyable aircraft. This feature does not affect any zoom; all it does is to define the focal point for every one of your views. To edit these values open the file "*bdg.txt*" and look for the following lines:

```
EYE_Z_POS_109
EYE_Z_POS_110
EYE_Z_POS_87
EYE_Z_POS_SPITFIRE
EYE_Z_HURRICANE
```

There seems to be no discernable unit of measurement, so trial-and-error will be required for the user to find the setting you want. There are also three other variables.

```
EYE_X_POS
EYE_Y_POS
EYE_Z_POS
```

These are the default values and if you do not have a specific aircraft value entered BoB will use these default pilot positions.

2.2.18 Infoline Configuration

The BDG added quite a few options in order to make the infolines more configurable; now you can configure both colour and alpha of the infoline text, it is also possible to deactivate its background. For the realism buffs, we added the option to suppress information about the current throttle setting, as well as information about the current ammunition status.

In order to change the text colour, edit the "INFOLINE_COLOUR" line in the *bdg.txt*. This value is again in hex format, but in contrast to the label-feature there is a setting for text-alpha (text opacity). A typical setting may be '7f00ff00' which means "display the text at half alpha ('7f' is '127' in hex) green ('00ff00' means "0 red, full green, 0 blue")". To toggle the infoline background, set the variable "SHOW_INFOLINE_BACKGROUND" to either "ON" or "OFF". Please note that message-lines (like your wingmen shouting "Shoot him!" – bet you've never heard that one) still get drawn in white with background

added. However, you can deactivate those messages altogether, please consult your game manual on how to do this.

You can also override the “Units” setting in the “View” configuration screen, so that German aircraft always display metric units while RAF aircraft feature imperial units. The relevant line in the *bdg.txt* is “USE_NATIONAL_UNITS”.

Activating the variable “HIDE_POWER_INFO” hides the information concerning your current throttle setting.

Historically, the only aircraft to carry an on-board ammunition counter in 1940 was the Bf-110. To simulate this, open the *bdg.txt* and look for the “HIDE_AMMO-COUNTER” line. Change the default value from “OFF” to “ON”, but note that this will remove any indication of how many bullets you have left, so the only way to tell that you are empty is when you squeeze the trigger and nothing happens!

2.2.19 Roving Cam

BoB now features a similar function as European Air War (or Strike Fighters) – a freely moving, (optionally) mouse controlled camera.

You can enter this “camera mode” by pressing **[ALT] [V]** (activating the autopilot first might be a good idea), and you can control the camera speed by pressing **[ALT] [G]** or **[ALT] [B]** (to speed up or slow down, respectively). Controlling the direction of movement is done by pressing the cursor keys on the keypad, if you want to stop at your current location (and reset the camera heading and pitch) press **[ALT] [F]**.

In order to control the camera with the mouse set the variable “MOUSE_ROVING_CAM” to “ON”. You can control the camera speed by clicking the left mouse button, or, if your mouse has a wheel, by turning the mouse-wheel after setting the ‘view zoom’ setting in the options to ‘mouse wheel’. This mode of controlling lets you go backwards as well!

2.2.20 UI Refresh

Windows XP seems not to be able to set a sane refresh rate on entering certain applications. This variable lets you specify a refresh rate in Hertz. Everything over 60 Hz should lessen the eye-strain.

2.2.21 Belly landing

You can now belly land your aircraft with the gear up on land or water if you keep your speed within 20% of the aircraft landing speed. You must ensure that the prop touches the ground first and not the wings.

2.2.22 Bullet dispersion

You must set the switch “Bullet_Dispersion” to “ON”. The default is “OFF”. The dispersion is 4 meters at 400 yards (max) for fighters and 16 meters for hand operated

guns (Max). The dispersion radius decreases as the range to the target decreases at the same rate (100 meters = 1 meter, 200 meters = 2 meters, 300 meters = 3 meters).

2.2.23 High Speed Buffeting

At speeds over 400 Mph the aircraft will buffet mildly. The buffet increases proportional to the speed increase.

2.2.24 Aero Surface Deflection

The amount of deflection of the Aero Surfaces is decreased as the speed of the aircraft increases. At normal and low speeds the aircraft aero surfaces will deflect normally (but less than the original BoB operation). Our hope and belief is that most users will find BoB must easier to fly with a smoother operation. At high speeds and during high speed combat the user will have to work harder with the joystick to get the same amount of Aero Surface deflection. At very high speeds the Aero Surfaces will deflect only a small percent so high speed dives are very dangerous.

2.2.25 Changed aircraft collision effects

Rather than having the aircraft blow up each time it collides with other objects and aircraft, other effects are now used (wings break, prop bends, fins break, or canopy breaks). This permits the user in some cases to return to base or at least bail out.

2.2.26 Frame-rate counter

We added a frame per seconds (FPS) counter that was already partially implemented by Rowan: it shows the current count, the average FPS, and the minimum and maximum FPS. Each time you toggle this display using the **CTRL F** key (or if you leave the 3d), the counter will be reset.

2.3 New Keys and their function

Keystroke	Function
S	Padlock enemy at the centre of the screen
CTRL S	Re-centres the padlock centre view
CTRL A	Autopilot toggle
ALT F6	Reverse Outside Padlock View
CTRL I	Toggles Other Aircraft View
ALT S	Multiplayer Suicide key
SHIFT F7	Decrease Game Time
SHIFT F8	Continuous in-cockpit zoom from min to max
SHIFT F9	Zoom to smallest Field of View setting
SHIFT F10	Zoom to middle Field of View setting
SHIFT F11	Zoom to large Field of View setting
SHIFT F12	Toggle between Field of View setting
CTRL F9	WinAmp music support -- previous track
CTRL F10	WinAmp music support -- Start music

CTRL	F11	WinAmp music support -- Stop music
CTRL	F12	WinAmp music support -- next track
CTRL	+	WinAmp music support -- increase volume
CTRL	-	WinAmp music support -- decrease volume
L		Look through floor window (<i>Stuka only</i>)
CTRL	F	Toggle (and reset) FPS counter
Q	W	Lean left and right keys are now sticky
ALT	V	Activates Roving Camera
ALT	G	Increases speed of Roving Camera
ALT	B	Decreases speed of Roving Camera
ALT	F	Stops camera and resets heading and pitch

Part III: The Campaign



3.1 Campaign Code Changes

A talented coder with the callsign of *El Fanto* delved into the campaign code and was able to make some basic changes to BoBs formidable campaign. We will proceed through these changes and discuss how they affect the campaign, under the hood.

3.1.1 Missing He111 Statistics

After you made an attack on that formation of He111s, the aircraft the player shot down were not visible in all statistics dialog window. This discrepancy is now corrected, its not "fog of war" that gave different numbers it was just poor counting by the computer.

3.1.2 Negative Aircraft Count

Occasionally a severely damaged flight threw the computer for a loop and caused the campaign to "lose" the twice. When this happened the number of available aircraft would fall below zero. To the player it would appear as if approximately 250 aircraft appeared out of nowhere for that squadron. This situation is corrected and the number of available aircraft will not go below zero.

3.1.3 Losing the Campaign due to Excessive Convoy Losses

One of the most frustrating problems with the campaign was the difficulty in getting through the convoy phase as the RAF commander. The basic code added points for allowing docks and convoys to be attacked and subtracted points if you shot down or damaged enemy aircraft and drove away the attack. The bug in this was if you allowed a dock to be damaged a point was added that day against your command, another point was added the next day, even if you stopped all the attacks because the computer saw the dock was still damaged. Therefore when you let a dock get damaged the computer continually counted that against you causing you to be sacked around the 4th of August. This fix changes the continually added value and now you only get penalized if you actually allow the dock to be damaged in that day. You can still get sacked if you always ignore the convoys and docks, but it should be more realistic now.

3.1.4 Wake Up Problem when Flying for a Specific Squadron

One of the options is to tell the campaign when you want to take over the flights, and this can be set to take over for one particular squadron. This procedure allows the player to fly a single life sort of campaign. The problem was if the player's squadron was assigned to intercept a raid, but was not the first squadron to be chosen then the campaign would never bring up the option to fly dialog box. El Fanto has now repaired this and the player will always be notified whenever the squadron is tasked.

3.1.5 Reworking of All Campaign Statistical Counters

The campaign keeps a double accounting system. The code keeps a counter for statistics used in the displays and another counter for the “decision engine”. Sometimes these counters would differ, causing the player to see react to one set of data while the campaign make decisions based on entirely different numbers. For example, a flight of Ju87s would appear on a raid when the dialog windows said that there were no such aircraft available. Often this was attributed to a “fog of war” effect, however, there is no such effect modelled in the campaign. Fixes to keep these counters in synch with each other should eliminate this bug.

3.2 RAF Fighter Tactics

The following sections are a look at how to use the 2D map to set and layout your campaign moves to maximize your chances to thrash the Hun above Britain's blue skies. The following deals with the deployment of basic tactical units – squadrons and flights.

Let's keep it simple and straightforward. Interception generally moves through the following phases.

- i. Readiness
- ii. Deployment
- iii. Engagement

Let's go through them one at a time.

3.2.1 Readiness

The single most important commodity you have in the campaign is not aeroplanes or even pilots it is time. A good commander knows ahead of all else how to manage time. The more time that elapses between the first detection of a hostile plot and your response getting airborne, the less chance you have of effective interception. The campaign models the RDF system available to you in 1940 pretty accurately, therefore, use it to your best advantage.

The player must manage their squadrons and pilots. Poor quality squadrons do a poor quality job. Use the readiness tools available to you and release squadrons whenever it is practical to do so, especially during periods of poor weather. Release your hardest pressed units, as even a few hours will improve their moral and condition. When you need them, they will be at their best if given the opportunity to rest. Try setting minimum “rest if” options in your “squadrons” menu or in the directives each time period. The squadrons will then be rested automatically if they fall below the minimums the player set.

Next, replace badly battered units, but do this with caution. As it was in reality fresh, and hence, inexperienced units will suffer correspondingly higher casualties in their first few combats. Use the directives menu to set standing patrols only over convoys or other high priority targets, otherwise do not use the programmed “responses” and cancel

automatic scrambling. If left to the computer the player may as well press the exit button now.

At the beginning of each day set a minimum number of squadrons to “at 2 minutes” using the squadron readiness tab. Depending on what stage of the campaign you are at, this should be a number that you are satisfied constitutes your minimum response. Too few will weaken your response, too many will tire your squadrons. Alternate the squadrons at 2 minutes readiness throughout the day so that the task does not always fall on the same units. Do not forget you will need both Spitfires and Hurricanes where possible, the rest of the squadrons should be left at 5 minutes readiness.

Responding to the raid requires time, so set the game non-raid speed at a level you are comfortable with. You can do this in the “Game speed” tab in the lower menu. Do not set it so high that events move so fast that you cannot control them. On this menu you can also set favourite squadrons, and take over options. The most important option is the take over “on raid forming”. You must know as soon as a raid is detected and react quickly. Set the time to pause on raid detection.

As time moves on during the day, try and anticipate events. If the weather is fine and no serious raids have come your way in the first two periods, it is likely that you will have busy tea-time. The AI plans its raids in 3 periods but raids may overlap.

One option is to get patrols up early. Patrols are a gamble, they may be a waste of time, but if you use common sense they may often be priceless. Standing patrols over convoys and coastal patrols are of limited value but are necessary, both to protect assets and keep your computerised senior officers happy. Don’t overdo them or they reduce your ability to react when serious threats do materialise. The above considerations fall into the “readiness” category. Once a raid is detected your chance to affect the battle due to management of squadron readiness is over.

3.2.2 Deployment

Once a raid or series of raids is detected, we now move into the “deployment” phase. The game should now automatically click into “normal” time, or paused depending on the setting the player chooses.

The raid marker shows the information available to you about the raid forming on the other side of the channel. Its raid number, strength, and height are recorded here in that order. One important note that the raid’s height is “at present”, and that this may well change. Historical raid markers also only show the head of the raid, that is, the position of the bombers. The fighters may or may not be present there, nor at the height indicated. Fighter information can be viewed, to do so you need to switch the markers to “show individual gruppes or staffels” in the “map filters” folder. Raid direction arrows may also be useful to you in determining where to setup your patrol areas.

While the game is in normal speed, take a few moments to assess the threat that this raid (or raids) represents, a lone recon aeroplane is easy to deal with. If you want to intercept it, and you should, click on the raid and then click “authorize” from the “hostiles” dialogue that is brought up. The AI will usually assign a flight of 3 aircraft automatically

from a suitable airfield. Stopping recon aircraft when possible should be a high priority, the Luftwaffe determines raid targets based on intel status to a large degree. Otherwise, study the information and gauge the threat accordingly. The raid will initially show as “forming” and no target data may be available yet. However, the relative position of the raid will give you an idea of the general target area. For example, a raid forming over Normandy is not likely to engage a target on the east coast. Nothing should be ruled out, but common sense should give you at least a fair idea of where it is headed.

The size and composition of the raid is the main precursor. Composition may show aircraft type and other information, such as if escorts are present or not. After gauging the degree of threat presented by the raid and having decided to intercept it, the time to deploy against it is now. On average, a raid will take between 15 and 30 minutes to form and reach typical targets in Southeast England. A patrol will average (depending on its readiness state) between 10 and 20 minutes to prepare, take off, and reach 15000ft within its sector. The time for the players patrol to reach an acceptable altitude is why time is so important. The time to decide the tactical details of the interception is now, not when your aircraft join combat. By then it is too late.

First, get your patrols up, section 3.3 has detailed instructions on how to do this. After determining, or guessing, where the raid is headed, place your patrols in a “holding” position close to the anticipated path of the raid, but not on it. These can be targets airfields, RDF stations, or preferably towns. Towns can be displayed by bringing up town and village markers up on the map from the “map filters” folder. The tactical positioning of your squadrons is important; too far forward and they may not be able to respond to changes of direction by the raid, too far back and they may not be able to intervene at the best moment. If the patrols are too far apart they cannot support each other, too close and they make a tempting target for the free escorts.

The key is not to think in terms of “horizontal lines”, but in terms of a chessboard, or a quadrillage, now expand that into a 3D chessboard. Most of the time aircraft travel in straight lines, but they are able to move more or less infinitely within this huge 3D chessboard.

The total number of aircraft to put up depends on the resources you have available, are prepared to commit, or the size of the raid. A large interception force may not necessarily produce better results than a smaller one, but will offer the enemy more targets. Generally, if you commit forces in the ratio of 1 : 3, that is a safe balance between under and over commitment. The overall quality of your force and your tactics will make a much more telling effect than sheer numbers. The mix again depends on what is available. You may already have committed forces to another raid and be short of Hurricanes or Spitfires, but ideally you need a balance. The Spitfires are superior performers to the Hurricanes, but the latter are better bomber killers. The Spitfire is an Air Superiority Fighter and is best suited to taking on the escorts.

Using paired squadrons is often the best policy, but beware that a large time penalty in deployment is attached to their use. Section 3.3 describes the time penalty for paired squadron deployment. The best use of paired squadrons is in anticipation where the time is available. The time needed to deploy wings, 3 to 5 squadrons, is rarely available. The patrol height depends on the height of the raid, try not to let the computer assign default altitudes (e.g. the one you assigned to the last patrol). Height is the most

important factor of a successful interception and is one the player can and should control for every patrol.

Typically, bombers will come in at around 15000ft, but this is not always the case. It is possible to have your intercepting fighters too high as well as too low. This depends on the relative position of the escort. If you place Hurricanes at 24000ft to intercept He111s at 15000ft and the escort is at 18000ft, your Hurricanes will become embroiled with the escorting fighters before they can get at the bombers.

Spitfires generally need to be higher, 18 – 20000ft is a good compromise. But beware, if there is a free escort assigned to “high escort” or “ceiling”, they may get “bounced”.

Assigning your Spits to a higher altitude to counter this possibility may result in them missing the raid altogether.

If you have more than one Spitfire squadron up, try setting different altitudes for them, for example at 18000 and 21000ft. This tactic will pay a dividend later during the actual interception. As you place each patrol, set the parameters under which you wish them to engage, set Hurricanes to prefer “bomber” targets and Spitfires to “fighters”. Set the tactics with caution, if you leave the default as “use Vics” your squadrons will attempt to use the old Fighter Command “Fighting Area Attacks” while remaining in tight vee formations. This is practically suicide if defending fighters are around, and not far short of it if they aren’t. “Dive and Zoom” is best used from high altitude but is of limited effectiveness on bombers because your fighters are spending so much time going up and down they put minimal lead on the target. “Line abreast” or “line astern” are extensions of the “Area Attacks” described above. These tactics will produce bomber kills with the drawback of reduced ability to respond to fighter interception. “Line astern”, however is useful in breaking up bomber formations since the aircraft will follow each other in rather than dissipate their attacks across a wide area. “Head on” tactics are extremely useful against bombers, but only if you get your squadron into a position from where such an attack is possible. If you fail to do this, the interception that relies on this method alone will fail. “Individual” is the most versatile method and about the only method that you can use against fighters.

As you assign a squadron “at 2 mins”, bring another to 5 minute standby to take its place. In this way, you maintain what all good commanders keep, and that is a reserve to deal with the next threat.

Once you have ordered your patrols into position, place the game back into accelerated mode at a moderate pace. The raid should have formed and begun moving into your airspace. Within a very short period of time events will have moved on to the point where it is not possible for you to get further units into the air in time to have an influence on this raid. Therefore, do not be tempted to waste assets if you find that the raid is bigger than at first thought. Use whatever assets you can, standing patrols in the vicinity or other assets allocated to another raid if necessary, but bringing up more units to engage this raid is now pointless and wasteful. The deployment period for this raid has now past.

3.3 Procedures for Campaign Tasks

The procedures section is a guide, not on the strategy for deploying your squadrons, but a “how-to” guide for making the campaign inputs you want.

3.3.1 RAF Patrols

This tutorial is designed to give the basics of setting up a patrol over a chosen destination. Patrols are a basic tool used to cover convoys early in the campaign; they can also be used as rapid response combat air patrols, or CAP, to respond to Luftwaffe raids. The one draw back to patrols is they have a limited time on station, or flying, and then the squadron must rest before going up again.

3.3.1.1 Patrols with Directives

First, let us look at the DIRECTIVES screen that is shown at the beginning of each time period. This screen lets you lay out broad orders to your controllers. The screen has several vital areas that directly effect which raids are intercepted, what is patrolled and by how many aircraft. The use of this screen is on of two ways to set patrols.



Figure 1. Directive Screen

Section 1 deals specifically with 11 Groups squadron, as you remember 11 Group is in the southeast of the island and endures the most of the fighting. Three options set patrols over different objectives. *Holes in radar* send aircraft to fly patrol when the Luftwaffe knocks out part of the radar coverage. *Over convoys* sets a constant patrol pattern over convoys as they come up the channel and into the harbor, with replacement squadrons scheduled to fly over the convoy as the previous squadron returns to base, or RTBs. This provides coverage of the convoy 100% of the time. One word of warning, check to see how many convoys are in the area. If you schedule six squadron and there are three convoys, then all three convoys will have coverage for as long as two squadrons are available, roughly two hours total. The possibility of leaving the convoy unprotected could then arise, remember allocate enough squadrons to cover all convoys in the area.

Coastal is different and gives you an interesting patrol pattern. If you call for six coastal patrols, you end up with six patrols at the same time flying over radar stations that are still active. These patrols are spread up and down the coast of merry ol'England. The one draw back to this is there are no follow-up patrols scheduled; this is a one-shot deal per period.

Section 2 deals with the size of patrol the player is sending out. *Squadron or less* implies that only one squadron, generally Hurricanes, will be assigned to fly the patrol. *Paired squadrons* have two squadrons, Hurricane and Spitfire, assigned to fly the patrol. The drawback to paired squadrons is that it takes longer to assemble and fly to the patrol, (or intercept). The squadrons' takeoff, then form on each other, and then fly to the patrol area. *Squadron or less* the squadrons takeoff and immediately fly to the assigned patrol area. *Big Wing* is an even larger formation that suffers from the same inherent problem of assembling the squadrons into the big wing before moving to the assigned patrol area. The advantage, of course, is that once that formation is in the air and a raid is spotted the player now has a powerful force to counter the raid.

3.3.1.2 Patrols set manually

The second method of initiating patrols is using a manual method. The manual method allows the player to set specific target that receive patrols, and are frequently used to have aircraft in the air ready to revector onto incoming raids.

Setting the patrol starts with the player deciding on what the patrol is to guard. I have chosen the Dover CH towers, just give a left-click and you will see the screen in Figure 2. This screen is also the INTEL screen accessed from the mission folder.



Figure 2. Location Intel Screen

Assigning a patrol is as simple as clicking on the *Patrol* button in the window, which brings up the screen in Figure 3, the Mission folder.



Figure 3. Mission Folder

The outlined line shows that patrol F001 is being flown by 32 Squadron with 12 aircraft. The patrol is currently on the ground readying aircraft and will fly to Dover CH for patrol. All of this can be modified without starting another patrol order by clicking on the Task button, which shows the screen in Figure 4, the task summary screen.



Figure 4. Task Summary

3.3.1.3 Patrol Modifications

The summary shows that currently one squadron of Hurricanes is going to fly a patrol over Dover CH. This screen allows you to change the number of aircraft assigned, but clicking on the *1 Squadron* line. In effect, you are making this patrol a *Paired Squadron* that was discussed earlier, with the penalties of a paired squadron. You can also change the squadron assigned, as shown above, from a Hurricane squadron to a Spitfire squadron if desired. For this tutorial, I changed the number of squadrons assigned to two. The player can then modify Hurricane orders by clicking on the Hurricane tab outlined. This will bring up Figure 5, the tasked Hurricane orders.

The Hurricane orders screen allows you to change three separate areas concerning Hurricane deployment, use, and unit. Figure 5 shows the preferred target of your assigned Hurricanes. I prefer to have the Hurricanes go after the bombers, if possible. This does not override the AI in defensive measures, although I cannot say with absolute certainty that with Bombers chosen the AI won't ignore fighters to concentrate on the bombers. Another problem is Me-110, with their ability to carry bombs, yet still be classed as a fighter. If the raid is all Me-110s this is not an issue, but if there are Me-109s and Me-110s then the possibility of the AI to become confused does exist.

Figure 6, shows the options for the Hurricane's attack method. I have not studied this extensively, except to know that Vic is not a formation to stay with is possible. Each attack method has an advantage depending on target type, the one problem with this

kind of control; you may inadvertently corner the AI into a bad decision with these options.



Figure 5. Tasked Hurricane order, Preferred Target Options



Figure 6. Tasked Hurricane order, Attack Method Options

Last, but by no means least is the ability to pick which Hurricane squadron you want to fly the patrol, click on the names squadron, outlined in red. You will then see Figure 7, the Hurricane squadron list.



Figure 7. Hurricane Squadron List

Choosing a squadron is as simple as double-clicking on the squadron you want to fly. One note of caution, this screen does not remove squadron who are currently unable to fly due to other restrictions. You can easily assign this patrol to a squadron that just returned from a sortie and is unable to fly for a period. A double-check of squadron status after leaving the orders area is a good practice. The easiest way to do this is from the Mission Folder screen, highlight the squadron assigned to the patrol and hit the *Squadron* button in the lower right. You will see Figure 8, the squadron readiness screen.



Figure 8. Squadron Readiness

This shows you the squadron is on the ground because the *required* line does not say *IN AIR*. It shows 16 aircraft available and they are ready to fly, from the *status* line.

All of these options, except squadron status, are duplicated under the Spitfire tab. The one thing to remember is if you have only one squadron assigned and it is a Hurricane squadron then the Spitfire screen will not work. You can visit the screen, but none of the options, *Attack Method*, *Preferred Target*, and *Squadron* will respond to mouse clicks.

As earlier stated, this tutorial increased the number of squadrons to two, one Hurricane and one Spitfire. After all the changes were completed, the new task summary screen looked like the one in Figure 9. The new task summary shows that a squadron of Hurricanes and a squadron of Spitfires will be flying patrol over Dover CH. One note, the campaign engine will try and bring squadrons from the same airfield as the first choice, so if you do not want both squadrons from Biggin Hill AF to patrol at the same time, remember to go to the aircraft tab and change squadrons.



Figure 9. New Task Summary

After closing all the windows, you will then see Figure 10, the final product. Squadrons 32 and 610 from Biggins Hill AF ready for an hour-long patrol over Dover, now jolly good luck then!

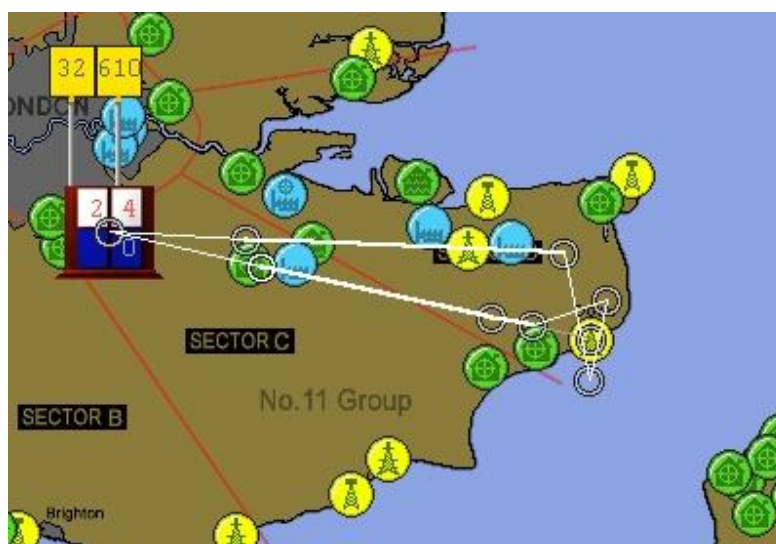


Figure 10. Patrol on Map Screen

3.3.2 RAF Route Manipulation

This tutorial will build upon the knowledge we gained from initiating a patrol, and move into manipulating the route that the campaign starts with.

First, I reaccomplished a patrol over Dover Chain Home station with a patrol from 43 Hurricane squadron out of Tangmere AF. The route information is displayed by opening the Mission Folder, highlighting the 43 Squadron line and clicking on *Route*. Figure 1 shows the procedure to reach the waypoint screen shown on Figure 2.

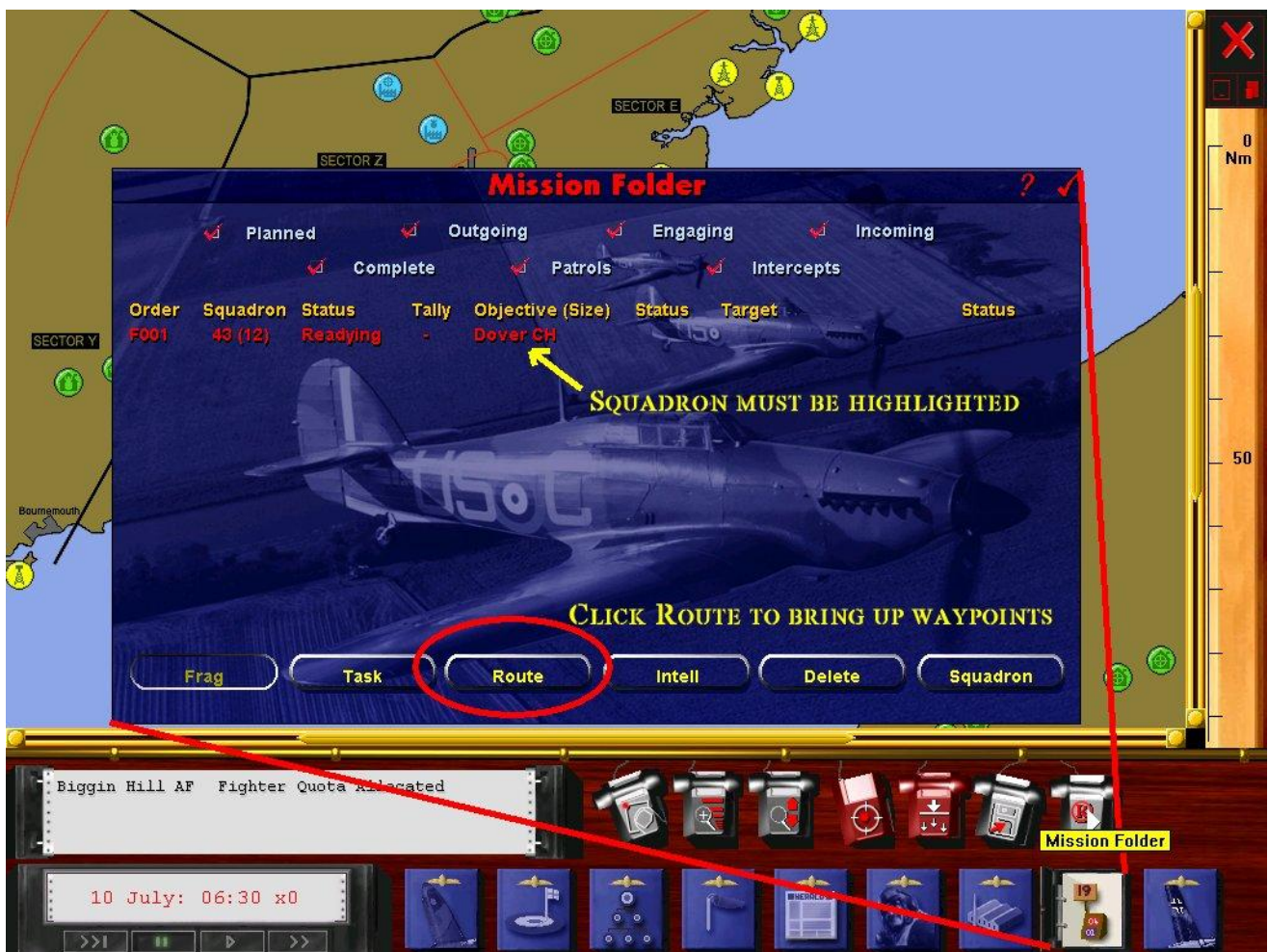


Figure 1. Process to reach Waypoint Screen



Figure 2. Route Waypoints

I have cut and pasted several images together to give all the information available. The player can move the mouse cursor over each waypoint to view the while labels designating what that waypoint is. The route screen has three tabs; takeoff, main route, and landing, which I have placed in the same picture for informational purposes.

Reviewing the information, we see that 43 squadron is taking off at 06:52 and will arrive at the initial point, or IP, 24 minutes later at 07:16. Last tutorial I mentioned that when you have paired squadrons or big wings, the aircraft had to takeoff and marshal before moving to the IP, which lengthened the response time. The waypoint labeled *Fighter Rendezvous* is the waypoint that those aircraft will circle until everyone assigned to that task arrives. This is why larger aircraft formations take longer to respond to a raid, they must all arrive at the waypoint and climb to 15,000ft before moving to intercept the raid.

Often the player will want a different time for his patrol, this allows you to sit down and plan an entire day's patrol. The player can assign four squadrons to patrol Dover CH and by manipulating the Time on Target, or ToT, for the IP of each patrol ensure four hours of continuous coverage.

There is one thing you can change at every waypoint and that is altitude, the only place to change time is at the IP of the route, double clicking on the circle that is *Waypoint: initial point* brings up the screen in Figure 3.



Figure 3. IP Waypoint Information

This screen, and the zoomed in map screen, give the player some valuable information. First, for those players who use the ground to navigate from takeoff to landing, you can see the IP is near a airfield for easy identification. If the player navigates like this, waypoint can and should, be moved to easily identifiable ground points. Second, the IP screen shows a lot of information.

Alt, ft:	Altitude in feet the waypoint is
ETA:	Estimated time of arrival at the waypoint in campaign time
REL to ToT:	Relative time to IP. If negative that means the waypoint should be reached BEFORE the IP
Fuel:	Time remaining before the tank runs dry, I can't confirm but I assume it is at a cruise setting for the throttle and not at 100% throttle. Battle will reduce this time
Escort Fuel:	Used for German bomber planning, gives you a time for battle over the IP
Bearing:	Compass direction from last waypoint, if you use manual navigation

The *Alt, ft* and *ETA* both have red arrows in their box, signifying that they can be changed. Figure 4 shows changes made to lower the altitude from 15,000ft to 14,000ft and change the IP arrival time to 08:00.



Figure 4. New IP Waypoint Information

Notice how the fuel time went from 51 minutes to 52 minutes, this is due to the fact that your not climbing to 15,000ft thereby saving your pilot some gas. Of course, if you change the time you arrive on station, you have to change the timing on the rest of the waypoints. Figure 5 shows you the new takeoff time to meet the 08:00 patrol time.



Figure 5. New Takeoff Time

The takeoff time has moved from 06:52 to 07:35 to meet your new IP requirements. The last item to change is the actual waypoint position itself. If you look at Figure 2 you can see the patrol area is 2-3 miles long in a north – south direction, not the most optimal direction to stop raids from France. Moving the waypoints is as easy as left-clicking on the waypoint and dragging it to its new location.

Words of warning, moving certain waypoints, most notably the IP, will cause all waypoints to change. While seemingly a random choice of the computer it is not. Moving the IP causes all other waypoints to be moved in relation to the *Waypoint: Estimated Location* waypoint and the new IP position. Think of it this way, when you move the IP, the target egress point and the patrol area need to move to still patrol over the target the player has selected. To overcome this move the IP first and then move the rest of the waypoints to the position desired. Figure 6 shows how I moved the patrol area to a better position.



Figure 6. New Waypoint Positions

There are the basics for manipulating the route for a task. Remember that this only changes the route for 43 Squadron. If you had a Spitfire squadron joining them, the player would have to go to the mission folder, highlight the Spitfire squadron and follow the same procedure. Otherwise, your Spitfires will arrive 30 minutes before your Hurricanes will.

3.3.3 Single Squadron Campaign

This section will explain how to setup a quasi-single pilot campaign. First, there are limitations to this method. The player is still responsible for the overall success of the campaign, in other words, you still have to either fill out the directives or assign missions yourself.

Basically, BoB has the ability to tailor which missions receive an interrupt asking the player if they want to fly. The default settings ask the player if they want to fly at every intercept, scramble, etc; but not at every take-off, ground attack, or landing.

Figure 1 shows how to access this from the main screen. You right-click down in the menu area, if you right-click in the map all you will see are *Zoom in/Zoom out*. After right-clicking, select *Time* and then *Game Speed*.



Figure 7. Main Screen Steps and Result

The *Accel* tab give the player the ability to control what speed the campaign is running at and what speed the campaign will jump to when certain conditions are met. So if you would like to react immediately to a raid forming, check the *Raid Forming* box and the campaign will move to real time when a raid is spotted on radar. This will allow you more time to set and modify your response.

Now to activate a “single-squadron” campaign you click on the *Fly* tab and you will see Figure 2.



Figure 8. Game Speed Fly Tab

The check marks show the default settings for the dialog box popping up asking you to fly. First, uncheck all the boxes in the All column, if any of these are checked it will override any options you choose in the Favorite column or *Favourite Units* boxes. There are many options you can follow here, if the player want to fly only Spitfires of any squadron just change the *Aircraft* box to *Spitfire*.

To fly a single squadron only, just left-click on the *Squadron* box and choose the squadron name you desire. Figure 3 shows the final result, of choosing to fly the 610 Spitfire squadron from take-off of every mission.



Figure 9. Flying 610 Squadron Only

At first there will be no change in the campaign. You will see missions generated and raids over France forming. You will be able to respond to all of these events normally, the only difference? You will not see a pop-up dialog window asking you if you would like to fly an intercept. When the 610 Squadron is assigned a mission you will see 610 squadron on the map shown in Figure 4.

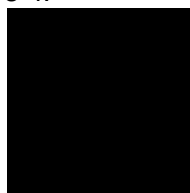


Figure 10. 610 Preparing to Fly

Although 610 Squadron is on the campaign screen and the player has not received a window asking to fly, DO NOT WORRY. What is happening is that 610's takeoff time has not arrived. If you looked at the squadron status it will show *Prepping to fly* and that the campaign time has not reached the takeoff time shown in the *Route* window.

When the takeoff time has been reached you will see the window in Figure 5.

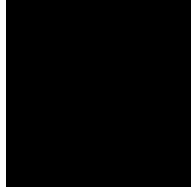


Figure 11. Taking over Dialog Box

Just click on the fly button and you will be sent to the cockpit awaiting takeoff. You fly the mission, land and when you exit you will go back to being Downing working a thankless job.

Its not quite single-pilot as you still have the campaign to worry about, but its pretty darn close.

Part IV: Developers View



4.1 Is that the sound of thunder or me shooting holes into that innocent cloud?

No, it is just the sound of that 20mm cannon from your Me-109. In the past if you had both guns and cannon selected you would hear both sounds, even if you were out of cannon shells. Now you will stop hearing the cannon sounds when you're out of shells. After the cannons are empty the only sound you will hear is the machine gun firing.

Sound files are as follow:

<i>gunlpfl.wav</i>	Luftwaffe AI and the replay
<i>gunoerlk.wav</i>	Luftwaffe player cannon sound
<i>gunrhbor.wav</i>	Luftwaffe player machine gun sound
<i>trainlo1.wav</i>	Luftwaffe player mixed firing sound

4.2 Understanding the *cardbase.rc* file

A more detailed description of the various options in the *cardbase.rc* file: Dave Whiteside from Rowan first posted this on SimHQ's BoB forum – thanks!

Why does BoB need a *cardbase.rc* file? The *cardbase.rc* file allows for the graphics engine to be tuned by hand and for specific machines and tastes. For example some of the W/Z buffer issues may be a matter of taste; if someone is happy to lose a small amount of resolution close up in order to gain a greater view distance then they can make tweaks for their own specific hardware. Also *cardbase.rc* allows bob to be more future proof.

For example when BoB was mastered the Geforce2MX card was not available, so anyone with that card would have had to have waited for a patch to make the card work. As it is they can just add the correct line to the *cardbase.rc* file and then get flying. Also a graphics chipset may be used by a number of card manufacturers to build their own cards some of which need different fixes. What we have attempted to provide is a generic set of solutions, but allow fine-tuning and tweaking where desirable.

Below is a slightly expanded explanation of what the different options within *cardbase.rc* do and why they are present.

General fixes:

VERTEX_FOG

Try this if there are fog problems (usually things are blue in the wrong places) or everything is blue. Some cards offer this ability as an option - for example the GeForce cards can switch off fog table emulation at which point this will need to be added to the fix list.

NO_TEXTURE_BLEND

If the sea looks wrong or the cloud layer looks wrong then this may be the option for you (also effect the dither ant aliasing). The sea animation will look poorer as will the cloud layer as only one texture will be applied to a polygon at a time.

NO_DEVICE_VALIDATION

This can fix the problems from the above in a more satisfactory manner. Also if the game can't even get into the 3D it is a good idea to try this one. What it really means is ignore the capabilities that DirectX reports and assume it can handle best guess approximations on its own.

MAX_TEXTURE_WIDTH <integer>

MAX_TEXTURE_HEIGHT <integer>

If textures seem to be cut off or you can't get into the 3D then using these two fixes with values of 256 or 128

NO_GUARDBAND_CLIP

Don't do guardband clipping if it is available. Some graphics cards allow for polygons to be drawn to a larger area than the screen buffer without clipping - they define a guardband region. This option should not really be needed unless a card enumerates wrongly.

STIPPLE_ALPHA

This can fix problems that may occur with the luminous transparency blending, use if the view reticule in the cockpit and the lens flares look wrong.

W_16BIT

Use if textures don't mesh or tear along polygon edges in the landscape and clouds. This means that during bob's polygon transform stage the whole size of the scene will be scaled down to prevent the sizes of polygons from being bigger than can be fitted into a 16bit value. It seems that some cards only do 16bit internally for texture calculations and require this.

Landscape texture issues:

Try these if tiles (squares) of the landscape are black or the wrong colours.

NO_TEXTURE_TARGET

This renders textures into the back buffer not directly into the texture. Can cause some stuttering.

SLOW_TEXTURE_DOWNLOAD

This doesn't use a hardware blit for copying the textures but a software copy. Can cause stuttering but only when lots of tiles need building (i.e. spinning the view around the horizon)

SLOWEST_TEXTURE_DOWNLOAD

As above, however only one of these fixes and the previous one can be active at the same time; if both are a line then the last to appear will be used. This is even slower than the above.

LOCK_TARG_FMT_TO_BACK_BUFF

As above, sometimes a better option as can be quicker, however can use up gfx card memory in 32 bit colour depth. However when the pixel formats are the same a hardware blit will often work. The comparative speed of using this and SLOW_TEXTURE_DOWNLOAD to fix non working blits is system dependant and affected by such things as AGP/PCI bus speeds and aperture sizes as well as hardware blit speeds.

Depth buffer issues:

If there is only a 16 bit or smaller z buffer available we will use a z buffer, otherwise we will use a w buffer if the card claims it can. The reason for this is that z is non-linear - it has a greater resolution when objects are up close to the viewpoint.

This is needed when there is only 16 bits available otherwise depth clipping of objects - such as hangars etc - against the landscape scenery may be unreliable. W buffering is however needed for detail in the distance, as it is linear all of the way to the horizon. This is why the far horizon option only works when a 32bit w buffer is in use - which requires a 32bit colour depth as the two are locked together on most hardware.

ALWAYS_USE_W

Always use a W buffer

ALWAYS_USE_Z

Always use a Z buffer. Symptoms for needing these fixes include the 3d not being displayed over the splash screen, only the 2d overlay.

FAKE_W_BUFFER

Fakes a w buffer using z, can prevent the above being used. This fixed fog and specula for the voodoo 5. In this case direct X thinks that a z buffer is in use but is fed w values in the z parameter for sorting.

4.3 AutoGen

AutoGen is a portion of the code that places objects like trees, houses, churches, etc, over certain parts of the landscape. This project is only at the beginning, but results are promising. Warning: some technical reading follows, not for the faint of heart!

4.3.1 AutoGen history and technical specifications

Very first, successful tests with automatically sprinkling of objects over the landscape have taken place. These prove that we know how to do the two most basic things needed:

At least sometimes, find out where cities and forests are so we can sprinkle objects there and actually telling the graphics engine to place a specific object at a place inside these areas. At first, we found out which parts of the terrain were "raised" by what amount and put things there, depending on the "raiseHeight". This cannot differentiate between the different areas that are not raised at all. So, now we try to use Rowan's "AreaType".

It will take some time to the first workable version, since there still is very much to do: We have not been able to understand or use all the "AreaTypes" yet. There seems to be an unknown, basic problem with (our understanding of) them.

- Sometimes objects will be generated in the water.
- From a certain horizontal or vertical distance onwards, we need to handle the objects differently.
- Objects on hill tops etc should sometimes be visible when the rest of the objects of the terraintile are not.
- We need special objects for AutoGen. The current houses have a LOD structure that is nice for their current use, but not AutoGen.
- Trees are done as billboards and again are quite nice in the standard BoB, but not in AutoGen
- Houses should be placed in street-like chains.

- Place objects in relation to the underlying texture.
- Optimize it for speed. This will probably mean pushing the code somewhere else with major changes.
- Do not place houses or objects on pathways on the border of cities.
- Finish the catalog of the existing objects.
- We need new textures below the AutoGenned objects. The object placement should match the textures.
- We cannot place groups.
- Object placement is not random enough.
- Only certain relative amounts of objects are possible
- Maybe create textures on the fly from the geometry? Blend it?
- *Maybe* implement additional stuff like power lines

It is very hard to predict how fast this system will be, but the ability to generate more realistic and varied terrain is in the future.

4.3.2 AutoGen implementation

AutoGen is implemented by adding information to the *bdg.txt* file found in the root Battle of Britain directory. A word of caution, this is not an exact science and there is the possibility that you will introduce errors into the game. Backup your *bdg.txt* file **before** starting any editing.

AutoGen quite simple takes 3D objects and places them in a specified terrain type. For example, you can tell AutoGen to place a sprinkling of Do-17s over the patchy grass terrain tiles. This is done by adding lines to the *bdg.txt* in the following format.

TEMP_AG_SN[AreaType,HowOften]=ShapeNum

Looks tricky doesn't it? Let's breakdown the formula into its components.

AreaType is the code that determines what type of terrain you want the 3D object placed. Each terrain type has an associated AreaType number with it. The following table lists available terrains and their AreaType numbers.

AreaType	Number	AreaType	Number
Darkgrass	0	Swamp	16
Mid Grass	1	Rough	17
Light Grass	2	Rough Grass	18
Garden	3	Patchy Grass	19
Golf Course	4	Sea	20
Field 1	5	Stream	21
Field 2	6	Village	22
Field 3	7	Dense Town	23
Field 4	8	Sparse Town	24
Field 5	9	Dense City	25
Spread 1	10	Sparse City	26
Spread 2	11	City Base	27

Spread 3	12	Sand	28
Spread 4	13	White Chalk	29
Spread 5	14	Air Field	30
Woods	15	Urban Buildup	31

ShapeNum is the Battle of Britain code for each 3D object in the game. If you look in the `\Docs\AutoGen\ShapeNum Listing.txt` you will find the list of all ShapeNum and their associated verbal description. Be warned, many of the verbal descriptions are vague and you may have to use trial and error to find out what they are. For example a tree in Battle of Britain is

Const ShapeNum TREE1 = (ShapeNum) 62;

So if you wanted to place the TREE1 object you would use 62 in the formula.

The last element needed is how many, how many of that object do you want placed on the AreaType you specified. There are four options for object density.

- 0 = few or 1.5% of the area for objects
- 1 = a bit or 4.6% of the area for objects
- 2 = many or 18.8% of the area for objects
- 3 = dense or 75% of the area for objects

So a formula that looks like this `TEMP_AG_SN[5,0]=62` would place the TREE1 object over 1.8% percent of every Field 1 area type that shows up on the computer screen in your pilots visual range.

Another command line for AutoGen that is used after all the AreaTypes and ShapeNum have been identified and entered into the *bdg.txt* file. That is the distance between each object as it is placed. This distance is measured in centimetres in the Battle of Britain world so we suggest using numbers over 2000 in the following formula.

TEMP_AG_DIST=2000

Below is an example that would be entered into your *bdg.txt* from Maususer, remember there will not be columns in the *bdg.txt*.

TEMP_AG_SN[0,2]=328	TEMP_AG_SN[14,0]=62
TEMP_AG_SN[1,0]=62	TEMP_AG_SN[15,3]=63
TEMP_AG_SN[2,0]=63	TEMP_AG_SN[16,0]=63
TEMP_AG_SN[3,0]=59	TEMP_AG_SN[17,2]=329
TEMP_AG_SN[4,0]=62	TEMP_AG_SN[18,0]=63
TEMP_AG_SN[5,2]=329	TEMP_AG_SN[19,2]=329
TEMP_AG_SN[6,2]=328	TEMP_AG_SN[22,3]=53
TEMP_AG_SN[7,0]=63	TEMP_AG_SN[23,3]=62
TEMP_AG_SN[8,0]=62	TEMP_AG_SN[24,3]=53
TEMP_AG_SN[9,0]=63	TEMP_AG_SN[25,1]=124
TEMP_AG_SN[10,0]=62	TEMP_AG_SN[26,1]=118
TEMP_AG_SN[11,0]=63	TEMP_AG_SN[27,1]=116
TEMP_AG_SN[12,0]=62	TEMP_AG_SN[28,0]=53
TEMP_AG_SN[13,2]=328	TEMP_AG_SN[29,1]=31

TEMP_AG_SN[30,0]=96
TEMP_AG_SN[31,3]=63
TEMP_AG_SN[31,2]=56

TEMP_AG_SN[31,1]=54
TEMP_AG_SN[31,0]=52
TEMP_AG_DIST=8000

4.4 BDG Code Bug Fixes

Some of the most important work the BDG coders did was to find and smash the bugs left in the program. For example; the 2D Crash to Desktop fix, the keyboard rudder fix or other changes to make BoB more stable. We decided not to go into great detail about all of these fixes, as they “automatically” will enhance your Battle of Britain experience. However, knowing that all flight simmers are detail-oriented, we have provided the following descriptions of these bug fixes.

4.4.1 BoB crash fixes

In the beginning...there were reports that BoB gets slower and slower after some time in 2D or that it crashes completely. Actually, MiG Alley has the same problem, especially Windows XP users complained about crashes. The problem is very bad, since your latest flights were lost and you had to redo everything from the last save.

At first it was thought that it is a memory leak, forcing Osram to implement a lot of debug code, send out a debug version, support the testers to get it to run with new ocx file, debug DLLs and an external debugger (dbmon.exe), and try to differentiate the "chaff" in the log files from real memory leaks. All the leaks he saw were minor and very hard to fix.

There were already a few hints it might not be memory, but GDI objects that were leaking. Pete Lilleyman made an important discovery when he saw BoB crashed when the number of GDI objects was 10000. A search on the internet showed that the number of GDI objects in Windows XP is not unlimited like Microsoft would like us to believe, but limited to 10000, or, more probably, 9999.

After this, Osram monitored the number of GDI objects on his own, Windows 2000 machine and found he could increase it by just moving the cursor about on the 2D. After some time it even crashed. This is how it normally is, the main job of the programmer is to reproduce it on his computer and find out under which circumstances it arises.

A look into the mouse-move function quickly revealed the bug. In coders language, a *GetDC* without a *ReleaseDC*. Once identified the fix was also fast.

One part of doing quality software is that in a situation like this you search for all "GetDCs" and look whether the same bug that was done in the mouse-move function was done in other parts as well. It turned out almost, but not all other GetDCs were ok.

4.4.2 Cloud Randomness

Clouds were not really random. BoB used the same cloud(s) all the time.

4.4.3 Keyboard Rudder Fix

One of the most important fixes for Osram was the keyboard rudder fix mentioned in the main part of this manual. He himself uses a keyboard, since his new computer does not like the pedals he has. This fix allows keyboard users the same rudder effectiveness as those flyers that have rudder inputs from their input devices.

4.4.4 Joystick Fix

There are also some errors you fix where you cannot completely say what effect it will have. For example, due to a copy and paste error, one of the DirectX joystick routines received an invalid parameter. Using a debug version of DirectX, DirectX complained about this and it was easy to fix this. Since the debug version knew it was invalid, did it ignore it? Does the release version of DirectX notice the error? If so, does it ignore the wrong parameter? If not, what effect does it have?

We do not know. We tested the joystick before and after the fix and see no difference, but it could for example be that the invalid value is passed on to the joystick driver and different drivers react differently to it, or that in non-tested circumstances it does have an effect. So you see, it is possible to find a bug and fix it, but do not know what effect, if any, it had.

4.4.5 Write Array Repair

Similarly, glop used a tool to automatically find a special form of bugs. An array is a list of a certain number, for example 20 values. When you write to the 21st value, some languages would notice the bug and warn you, but a "low level" language like C(++) does not and simply writes the value *somewhere*, overwriting god knows what with effects that are almost impossible to predict, but if you see the 21st element of 20 accessed, then you know it is a bug and you will hopefully know what element you really want and you will be able to fix it. While you cannot say "I have fixed the wrong behaviour xyz", you have increased the quality of the software.

4.4.6 Error message output

Output of error messages was broken during the 3D display. Additionally, the output of error messages was completely broken under any circumstance for Windows 2000. The fix will not directly help people as it will not avert errors, but when there is an error, we might know the reason when we might not with broken error messages. So, we might be able to do something where we would only scratch our head otherwise. Also, it can help to find errors earlier.

4.4.7 Panning bug

There was a bug that affected the pan speed left-right and right-left in different ways. So, one direction was noticeably faster than the other.

4.4.8 Gun and Cannon sounds

Wrong gun sounds were played with some Luftwaffe planes, in addition to the cannon sound playing even when the aircraft was out of cannon shells.

4.4.9 Flap buffet fix

The flap buffet sound was being played when on the ground with the flaps down, which is obviously wrong. The code was corrected to stop this annoying little bug.

4.4.10 Winds aloft correction

If there is a constant wind, especially a strong one at higher altitudes, Rowan let the bullets drift from the normal path. One problem, however, you have to understand that the airplane and the bullets "swim" in the same, constant wind and *relative to the airplane* the bullets behave as in zero wind. If you are sceptical, believe Osram, who has a master's degree in theoretical physics and worked on an expansion of the general theory of relativity. This physics error has been corrected in the BDG patch.

4.4.11 Ground turn or Radio turn bug

One very funny bug was when you played RAF in the low level attack. When you were still on the airfield and padlocked the approaching Me110s and told your comrades about them, they would try to do an aerobatic manoeuvre to escape the danger. This looked quite a lot like synchronized swimmers. It was clear what to do about this, you may not do aerobatics while on the ground and Osram also quickly found the place in the code where to add it, but it was harder to know whether you are on the ground or in the air.

You have to see that we have to understand a 600000 lines monster by ourselves. A software-engineering professor told me that this might well be impossible. Well, Retro found out how to check for being on the ground and so finally fixed the bug. It seems the fix in 0.90 was only half hearted; we hope to have finally fixed it in 0.95.

4.4.12 Fixed WK_LANDSCAPE 1 / Stutter fix

Fixed the wrong tiles with the WK_LANDSCAPE=1' setting!!

This can be seen as **the** stutter fix, since this mode fixes the biggest stutter cause in BoB, but until now lead to visual artefacts.

4.4.13 Stuka-grunt-bug fix

Sometimes you damaged an airplane and the pilot wanted to bail out, but could not because the cockpit was too damaged to open. In this case, the AI did not know what to do and while being inverted kept pushing the stick forward to easily fall free of the airplane. Thus the plane "skyrocketed" while the persons aboard grunted. Since it happened most often and was first observed in the Stuka, it was called the "Stuka grunt bug".

4.4.14 Pampa bug

This one was fixed by preventing a "division by zero" error. For zero wind the bug meant that the aircraft was placed in the "pampa" and you only see uniform blue above and green below and then it would CTD. This bug had been there since the original version.

4.4.15 Videos in Windows XP

...work again, thanks to Jamm0r.

4.4.16 Wrong files in 0.90 meant WinAmp did not work

This time we ship the correct keyb3d.bin, which will make WinAmp work. Also, in 0.90 there were problems with the language files.

4.4.17 The "no more hat panning after mouse pan" problem

...was fixed by Hyper-.

4.4.18 Many bug fixes/improvements in robustness in 0.95

There are simply too many to have a chapter for each or even to have a complete listing here:

- Retro fixed the screenshot modes.
- made the (non-functioning) 'Reset Ground Target' work
- Some damage effects (especially the "parachute with chain") used gusts. If they were switched off, then there was no damage.
- fixed a bug with error output that lead to a CTD when the *bdg.txt* contained a wrong keyword.
- Normally, the BDG output is cached to improve performance. When BoB crashes, the cached information is not written to disc. There is now a *bdg.txt* variable "BDG_OUT_CACHING" which lets you disable caching in order to debug CTDs.
- Changed the default MIN_FRAME_TIME_MS value. This only affects people that do not have a *cardbase.rc* file yet.
- Added Debug stuff for the **ALT** **X** crash, activated by the *bdg.txt* line "DEBUG_STUTTER=2048"
- "behind the scenes stuff": better error handling and messages, check for more error conditions etc (stub3d, lib3d etc), fixes in debug code, fixing the *bdg_out.txt* debug output that resulted in information loss, more *bdg_out.txt* debug output, additional debug features, checking the version of the ocx files, making things more robust.

0.95 contains some new code that could also lead to new problems and we did not have the time to test it in depth as we would like before the release, but the above measures mean that in the medium term, after more testing and use it should definitely be more stable than ever.

4.4.19 Fuelcock Fix

A non initialized variable caused the inability of setting the fuel cock in the Me109 and Hurricane. This has been fixed.

Part V:

Appendices

5.1 Battle of Britain Development Group History

After a long series of flight simulations for home computers and PCs, "Battle of Britain" is Rowan's last PC flight simulation. Since some people wanted more support from Rowan, Plummerx initiated an online petition, which still can be found online at <http://www.petitiononline.com/BoB/petition.html> . The result was a last, final patch from Rowan and we were told we might get the source code.

The BDG was born on 30th July 2001 when Ken Cook from SimHQ invited a band of BoB devotees to join a private forum to aid the development of Rowan's Battle of Britain, which had been released early in 2001. The forum was set up in the knowledge that it was probable that Rowan would release the source code for BoB into the public domain. In an email sent to Paul "Bader" Rogers on June 13th, 2001, David Pringle of Empire Interactive suggested that the source code for BoB could be released.

What followed was a protracted period of discussion and disbelief culminating in the release of the code on September 26th, 2001, in that time the BDG had developed methods for handling bug reports courtesy of Murphy and planned many of the improvements they intend to add to the game as well as prioritizing the bugs to be fixed.

Although some difficulties had to be overcome, the executable itself was successfully built on the same day by Osram, although with kludges and without some auxiliary files. Quite a lot of people were interested and saw whether they could compile the 600 000 line monster as well.

The first improvement that was tried was the Me109 range bug. It was thought that it was a simple matter of reducing the Me109 fuel load. After it was found out that it probably is not just a problem of the initial amount or the usage, but a problem with the transition between 2D and 3D, this fizzled out.

However, it turned out to be easy to enable other zoom ranges (field of view). On October 6th, Osram published a test version with zooming inside the BDG. It quickly turned out that a larger zoom range works, but has some bad side effects. These could mostly be overcome. A period of intensive testing, sometimes with one or more new executables each day followed.

A programmer that works without a version control system, especially on a program this size and with several people working on it, shoots himself into the knee. Unfortunately, opposite to what we had hoped for, the license was not GPL compatible. While it is fair, it cut off many things for us, for example getting the services of <http://www.sourceforge.net> .

We did not succeed in changing it. So, we had to roll our own CVS server and it was a major milestone as end October/start November the CVS server was up. Osram administers it, with some initial help from Weasel75 and glop.

After several iterations it was time to release a BDG beta to the public. This was done on the November 10th, 2001. It contained quite a lot of features that were easiest to implement, but that people wanted. Before the release, there was a time of frantic testing activity and we took on some new testers.

After the release of the source code for BoB there was a suggestion that the code could also be released for MiG Alley, another Rowan creation, many doubted that this was likely, but on November 8th, 2001 Rowan released the code for MiG Alley as well. At this point the decision was taken to try to address some of the issues with this game within the BDG as well. The reason for this decision was that there is significant commonality within the code for the two games and that there is probably not enough manpower for a separate effort. This also slightly swelled the numbers of official BDG members.

The next important release for the BDG was Rowan's release of the toolset needed to successfully modify BoB's artwork. After long waiting, this was made available on the December 19th, 2001. Also Rowan included the source artwork for the Spitfire, it took some time just to "compile" the Spitfire back into a form BoB can use, even without a transparent prop, let alone generate a new 3D file for BoB.

On 1.2.2002 the first version of the BDG web site went public, with a poll, several news stories, and a new "mod" by Bader, was created by RSColonel_131st.

In order to move forward the BDG will need as much support as they can get, in research effort into historical accuracy, actual coding experience, model generation, testing, test reporting and general good ideas. Anyone who feels to have the experience and/or enthusiasm to help should leave a note in our forum on <http://www.bob-ma.org> .

The BDG would like to thank SimHQ for hosting the forums we use to coordinate all our work. All of this would not have been possible without Plummerx, who raised a petition, but also Rowan who did something that only very few commercial software producers do, and set their brainchild free.

5.2 Battle of Britain Development Group

The people who brought you this version, listed in alphabetic order.

AtariBaby

ENGINE_ALWAYS_RUNNING feature.

BOBC

Artist: Textures

Carlos "Pipo" Cuevas

Translation, coding.

Lukas "Retro" Friembichler

Programmer: (Some) UI, AI, views, tools, bugs

Siegfried "El Fanto" Hempfer

Programmer: Campaign fixes (far from complete, but it's a beginning), Vehicle Manager.

Simon "eko" Knott

Tester, Artist: 3d models, textures

Wolfram "Osram" Kuss

Programmer: works on 2d, 3d, quickmissions, tools, pays for the website ;)

Buddye "Buddye1" Lavender

Programmer: Belly landing , Bullet dispersion effect, High Speed Buffeting, Aero Surface Deflection, A/C collision effects, and hopefully more realistic bullets that include drag and gravity effects

John "Maususer" C. Maleski

Tester, Artist: Textures, landscape, 3d models

Welsey "Wildman" Netcher

Manual

Parano

Programmer: MiG Alley BDG UI

Redband

Tester, Artist: Textures

Paul "Bader" Rogers

Tester, Artist: Textures, landscape, artwork pack, Missions

Jeffrey "stickman" Stickley

Tester, Artist: Textures, 'How-to' guides

Miro "Jammer" Torrielli

Programmer: D3D9 rasterizer, new weather / sky effects + AI, historic campaign weather, XP Video fixes, and MSDEV7 compatibility, GPL consultant

KH "viper-" Zeller

Programmer: BIN's , HiRes Cockpits, Multiskins mods

Testers:*James "JamesB" Bunn**Old Dux**JRT**Benny T Jorgensen**Stefan "SaimOn" Simon***(Our shiny new) Website:***GT***Localization***Jammer, Osram, Retro**Jose Luis "luke97" Galvan***Special Thanks:***Chris "Not Enough Red"**Dave "Rowan" Whiteside**Ken "KC23" Cook*

People who worked on previous versions:

RSColonel_131st	Original web site
Tailslide, Dragonflyer, Weasel75, Glop	Coding and help with CVS
Eric Mantelet	French localization
Rattler6, Murphy, Squiffy	Beta Testers
Lobo_Mau	3d modelling
Jiri "hyper-" Fojtasek	Coder
Julian "Codec" Onions	Flight model, vehicle manager foundations
Captain Farrel	Texture templates
Ursus	HiRes cockpits
Mark "A.Fokker" Linnemann	MP host
Janders	Some of the screenshots in this manual
Carlos 'PIPO' Cuevas	Translate Manual to Spanish
Fuelcock	Realism researcher

We are still looking for someone to help with the Italian and Russian versions. If you have that skill and are interested in helping please go to the BDG web site and contact us.

5.3 Acknowledgements

BoB uses the Expat XML Parser version 1.95.6 available from <http://expat.sourceforge.net/> .

5.4 3rd Party Modifications

BDG-Web, Home of the BoB Development Group
<http://www.bob-ma.org>

Eko:
<http://www.eko.clara.net/bob.html>

Maususer:
Clouds, Grass, 109 schemes
http://www-personal.umich.edu/~maususer/Battle_of_Britain_stuff.htm

Plummerx:
A Tribute to the Few. All round good fun plus the BoB theme!
<http://www.domin8air.com/bob/enter.htm>

Retro:
Most of the BDG stuff.
<http://stud3.tuwien.ac.at/~e9826217/BOB/index.html>

SimHQ:
<http://www.simhq.com>

SimHQ Message Boards:
<http://www.simhq.com/cgi-bin/ultimatebb.cgi>

Stickman:
<http://home.hawaii.rr.com/vadenstick/>

The Soupdragon:
Radio Sounds and some videos to drop into the game, including a wonderful 60sec Intro sampled from the 1969 film.
www.ctsprods.f9.co.uk

Viper-:
The best Visual effects: Smoke, Smoking Tracers, Explosions, File and Skin Manager and new cockpits. Lots to choose from.
<http://wbviper-.bei.t-online.de/vpbob.htm>

5.5 References and Historical Sources

The Battle of Britain - Official RAF Site.
<http://www.raf.mod.uk/bob1940/index.html>

Battle of Britain Source Code

<http://www.simhq.com/simhq3/sims/downloads/bob.shtml>

Rowan Software's Official Homepage:

<http://www.rowansoftware.com/rowan/welcome.htm>

5.6 Artwork Notes.

Hawker Hurricane 'Multiskin' Mod for Rowan's Battle of Britain.

Very special thanks to:

-Viper- for making this mod possible with extensive 'bin' modifications. This is incredibly painstaking work that only he understands. Ausgezeichnet, KH!!

-Cpt Farrel for making his excellent templates available, upon which these mods are based. An excellent start to work from!

This update introduces historically accurate automatic selection of paintschemes and markings for all Hurricane Squadrons in the period July-October 1940:

Some comments on RAF Fighter command paintschemes 1939-1941.

The colours Dark Earth and Dark Green were standard for upper surfaces until May 1942. There was little variation in the patterns other than a mirror-image application. The two types were designated 'A' and 'B' schemes. ('B' schemes are modelled here).

Undersurfaces and underside national markings were subject to a number of changes:

-In early 1940, the official scheme had a white right wing, black left wing and split fuselage colours to match. In fact, the manufacturers interpreted this as requiring an aluminium finish to the fuselage, which was an error, but it was not corrected for a long while. No underwing roundels. (Before this date various versions with and without roundels had been used). The split colour schemes were to aid ground identification.

-On 18th May 1940 Roundels were reintroduced, but with a yellow roundel ring on the left wing. The entire fuselage and tail surfaces were now split white/black for new aircraft. Black was intended to be the specific colour 'night', though this did not always happen.

-On 6th June 1940 the colour designated "Type S" was introduced for undersurfaces, without underwing roundels. With modern perspective (and communications methods), it is easy to be critical of the problem that neither manufacturers nor existing squadrons actually knew what "Type S" looked like! More than one written clarification was made, but the effect was to introduce a range of colours in the field from 'Duck Egg Blue' to 'Duck Egg Green' to 'Light Grey'. To compound the problem, official stocks of "Type S" paint were not available freely until over a month later than this. Each squadron/maintenance unit and each manufacturing location had to interpret what can best be described as 'imprecise' instructions. I have modelled a common 'Duck Egg Bluish Green', which is indeed how one Ministry correction described the colour. (Later on, the Ministry would correct the description to be 'Sky', but by then the term 'Duck Egg Blue' was popularly established.)

- On 1st August large (45/50") roundels were reintroduced.
- After the official end to the battle, from 27th November 1940 a black left wing was reapplied, along with a sky spinner and tailband.

As some of these changes were introduced in the middle of the Battle of Britain there was considerable field variation, as one might imagine.

Upper surfaces and markings did not vary within a Squadron during the Battle all that much, so these are automatically selected by squadron. Undersurfaces changed with time.

Two styles of fuselage roundel (35" and 49") are modelled as well as three fin flash types (Tall, Short, Full Tail covering).

Nearly all have been confirmed from photographs. I have found sources for 30/34 Hurricane Squadrons. Where gaps existed the RAF's official Battle of Britain site was used. www.raf.mod.uk/history/cam_hist.html

Note: historical purists may spot some concessions to gameplay here, since Squadron-wide markings were rare in the RAF at this time and usually badges/motifs were carried by individual planes only. I hope you forgive the use of some markings more widely than in real life.

Sqdrn	Type	Code	Fuselage	Fin	Flash	Underwing	Underwing	Spinner	Special
1(C)	Hurricane	YO	49"	Tall	none	Sky	Black		none
1	Hurricane	JX	35"	Short	large	Sky	Black+ yellow ring		Wasp
3	Hurricane	QO	35"	Short	large	Sky	Black		none
17	Hurricane	YB	49"	Full	none	Sky	Yellow	Flying	Popeye
32	Hurricane	GZ	35"	Thin	none	Sky	Black		none
43	Hurricane	FT	35"	Short	large	Sky	Black		none
46	Hurricane	PO	35"	Full	large	Sky	Black		none
56	Hurricane	US	49"	Full	none	Sky	Black		none
73	Hurricane	TP	35"	Full	large	Sky	Black		none
79	Hurricane	NV	35"	Full	none	Sky	Black		none
85	Hurricane	VY	35"	Full	none	Sky	Black	Hexagon below cockpit	
87	Hurricane	LK	49"	Full	none	Sky	Black		none
111	Hurricane	JU	35"	Thin	large	Bl./Wh.	Black		none
145	Hurricane	SO	49"	Full	none	Sky	Black		none
151	Hurricane	DZ	35"	Thin	large	Sky	Black		none
213	Hurricane	AK	35"	Short	large	Sky	Black		none
229	Hurricane	KE	49"	Thin	large	Bl./Wh.	Black		Penguin
232	Hurricane	EF	35"	Short	large	Sky	Black		none
238	Hurricane	VK	49"	Short	none	Sky	Black		none
242	Hurricane	LE	35"	Short	large	Sky	Black	Hitler emblem	
245	Hurricane	DX	35"	Short	large	Sky	Black		none
249	Hurricane	GN	49"	Full	large	Sky	Black		none

253	Hurricane	SW	49"	Full	large	Bl./Wh.	Black	none
257	Hurricane	DT	35"	Short	large	Sky	Black	none
263	Hurricane	HE	49"	Tall	large	Sky	Black	none
302(Pl.)	Hurr	WX	35"	Thin	large	Sky	Black	Polish National
colours								
303(Pl.)	Hurr	RF	35"	Thin	large	Sky	Black	Polish National
colours								
310(Cz.)	Hurr	NN	35"	Full	large	Sky	Black	Czech National
colours								
501	Hurricane	SD	35"	Full	large	Bl./Wh.	Black	none
504	Hurricane	TM	49"	Thin	large	Sky	Black	none
601	Hurricane	UF	49"	Full	large	Sky	Black	none
605	Hurricane	UP	35"	Thin	large	Sky	Black	none
607	Hurricane	AF	35"	Full	large	Sky	Black	none
615	Hurricane	KW	35"	Full	large	Sky	Black	none

This mod is open to improvement as there are possibly errors/improvements. This is the first time this has been attempted in a computer simulation and the process of research has highlighted the inconsistencies of information available.

Bf109E 'Multiskin' Mod for Rowan's Battle of Britain.

Very special thanks to:

- Viper- for making this mod possible with extensive 'bin' modifications. (Still an art only he understands!)
- BOBC for providing 90% of the work on the skins. Thanks to both also for historical research.
- Vadenstick for additional help with textures and with markings/emblems research.
- Maususer for providing the basis for some reworked propellers.
- William Marshall for allowing us to use some of his Luftwaffe emblems.

This mod introduces automatic selection of three things for Messerschmitt Bf109E units:

- 1) Aircraft Paintschemes ("skins")
- 2) Unit Emblems
- 3) Officer and Staffel identification markings.

These are explained in the section below.

1) There is now automatically selected variability in PAINTSCHEMES for the Bf109s in the BoB period.

In real life these were notoriously mixed and whilst it is not possible to model every single plane or indeed Staffel, this mod does introduce the following representative types:

Camouflage scheme 1: Colours RLM 71/02/65. Full Yellow cowling and tail.

JG 2 I/II/III

JG 3 I/II

Camouflage scheme 2: Colours RLM 71/02/65. Camouflaged nose and tail.

JG 3 III

JG 26 I/II

Camouflage scheme 3: Colours RLM 71/02/65. Partial Yellow cowling, camouflaged tail.

JG 26 III

JG 27 I/II/III

JG 51 I/II/III

JG 52 I/II

Camouflage scheme 4: Colours RLM 71/02/65. Camouflaged nose and tail. Light mottle on fuselage. (based on a II/JG54 scheme)

JG 52 III

JG 53 I/II/III

JG 54 I/II/III

2) It also introduces historically correct UNIT MARKINGS. The aircraft carry Gruppe/Geschwade emblems and many carry Staffel insignia. There are even some special unit commander logos modelled on individual planes. The layout of the 'emblem' file is as follows, so you can see for yourself which units carried a specific badge (if they did and we have modelled it, an X is shown):

	JG_logo	Grp.I	Grp.II	Grp.III	Staffel:1,2,3,4,5,6,7,8,9	Kommodore
jg2	x			x	x x x x	
jg3	x	x	x	x	x x x x	
jg26	x				x x x x x x x x x	x (Adolph Galland)
jg27		x	x	x		
jg51	x	x	x	x		
jg52	x	x		x	x x x x	
jg53	x					
jg54	x	x	x	x	x x	

3) In addition, JGKom, JGAdj, Techn Off, GKom, GAdj are also correctly represented with specific markings
(Chevrons/slashes/bars)

5.7 How to do hires screenshots

- You need irfanview. if you do not have it yet, download it for free from <http://irfanview.tuwien.ac.at> .
- You have to set two variables in bdg.txt with an ascii editor:
Set BOB_SCREENSHOT_MODE to 3. The possible modes are 0 = "normal" stills, 1 = video, 2 = panoramic 3 = tiled (large to huge screenies) 4 = lenticular.
Set NO_OF_HIRES_SCREENIE_TILES to the number of tiles across. Say you run 1024 x 768 and choose "4". Then you will actually get 16 tiles and the complete screenie will be 4096 x 3072 pixels (4*1024 x 4*768).
- I suggest removing all screenies from "stills" so you know which ones are new.
- Run BoB and switch off everything that you do not want to be in each of the tiles like art. horizon ("h"), EPI (BDG/game stiings), infoline ("i").
- Go into pause to freeze movement and propellor
- Choose FoV and view direction
- Press "print screen". See the image "zoom in". Press printscreen altogether NO_OF_HIRES_SCREENIE_TILES*NO_OF_HIRES_SCREENIE_TILES times in slow succesion. It will "zoom out" again afterwards (at least once).
- For a first test, you can skip this step.
Unfortunately, there is a one pixel broken line all around the picture. To remove, go into IrfanView/File/Batch conversion/rename.
Add all the pictures and go into "set advanced options". Activate crop. As left, input "1", as widht 2 less than the total width.
Example: You have 1024 before, so you want a new width of 1022. Manage Height in the same way. Click "ok" and in the dialog press the Start button. This will convert the pictures, removing the bad lines. As output format I choose also BMP, since it is lossless (don't use JPG or so).

- have a look at the tiles and understand how they are laid out. For me, BoB decided to start with image shot_039.
The screenies start at the top right corner of the complete view and go down.

For example:

right most column, top to bottom: 39..46
next column: 47..54
next column: 55..62
etc
left most column: 95..102

- Go into Irfan view.
Load image 39. Say "image/create panorama/vertical" and add image 40 to 46. You can do this via MultiSelect. Save as "new39.bmp".
Load image 47. Say "image/create panorama/vertical" and add image 48 to 54. Save as "new47.bmp".
etc
- When done, you have loaded new95.bmp. Say "image/create Panorama/horizontal" and add new87, new79 etc to new39.
You now have (in the example) a 110MB, 7k x 5 k pixel image :-).